Portfolio

Jonghwan Shin

2026. 11. 02

Projects

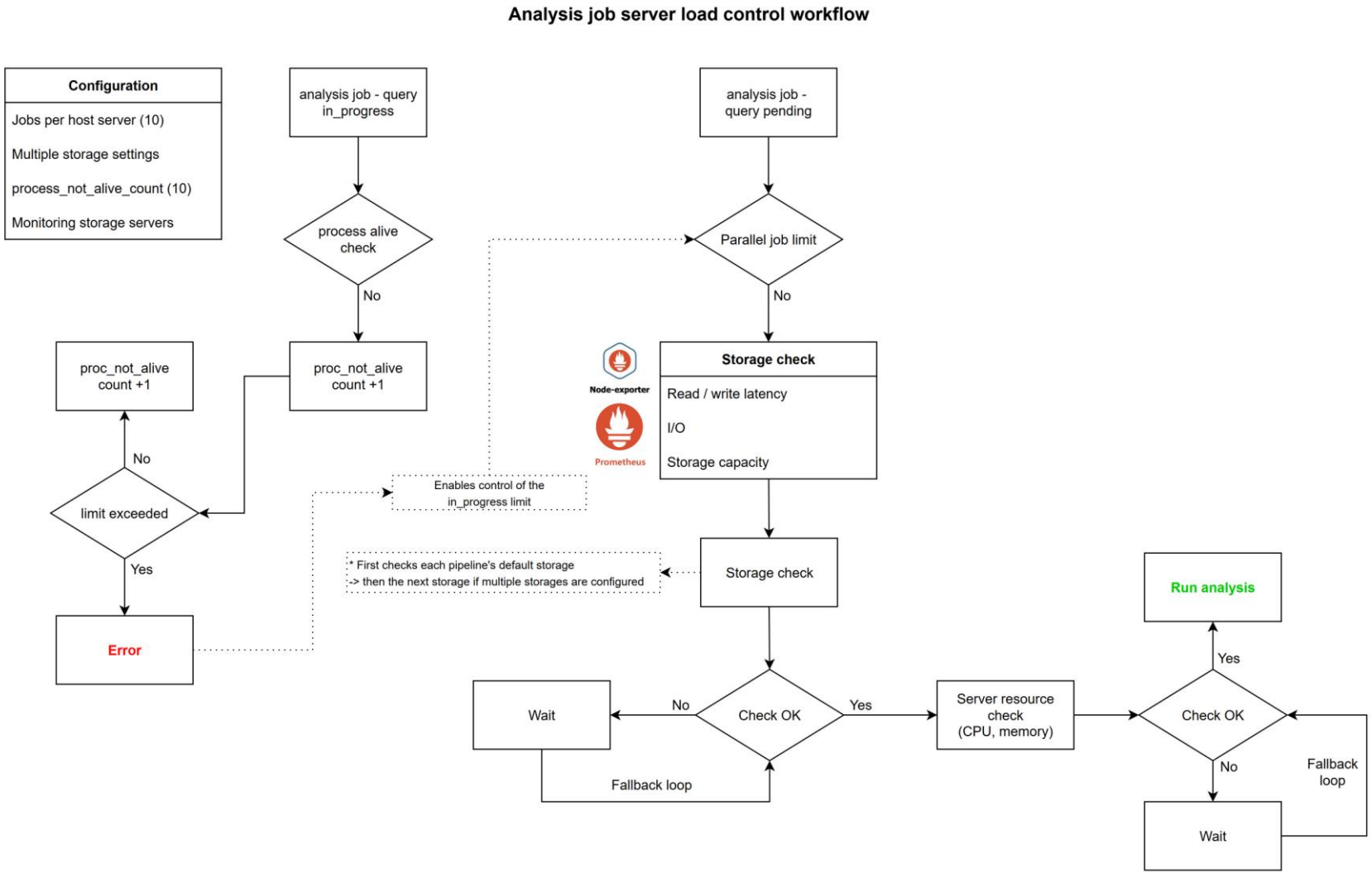
1. Analysis Automation Web Service (migration to Kubernetes)

2. CI/CD Pipeline

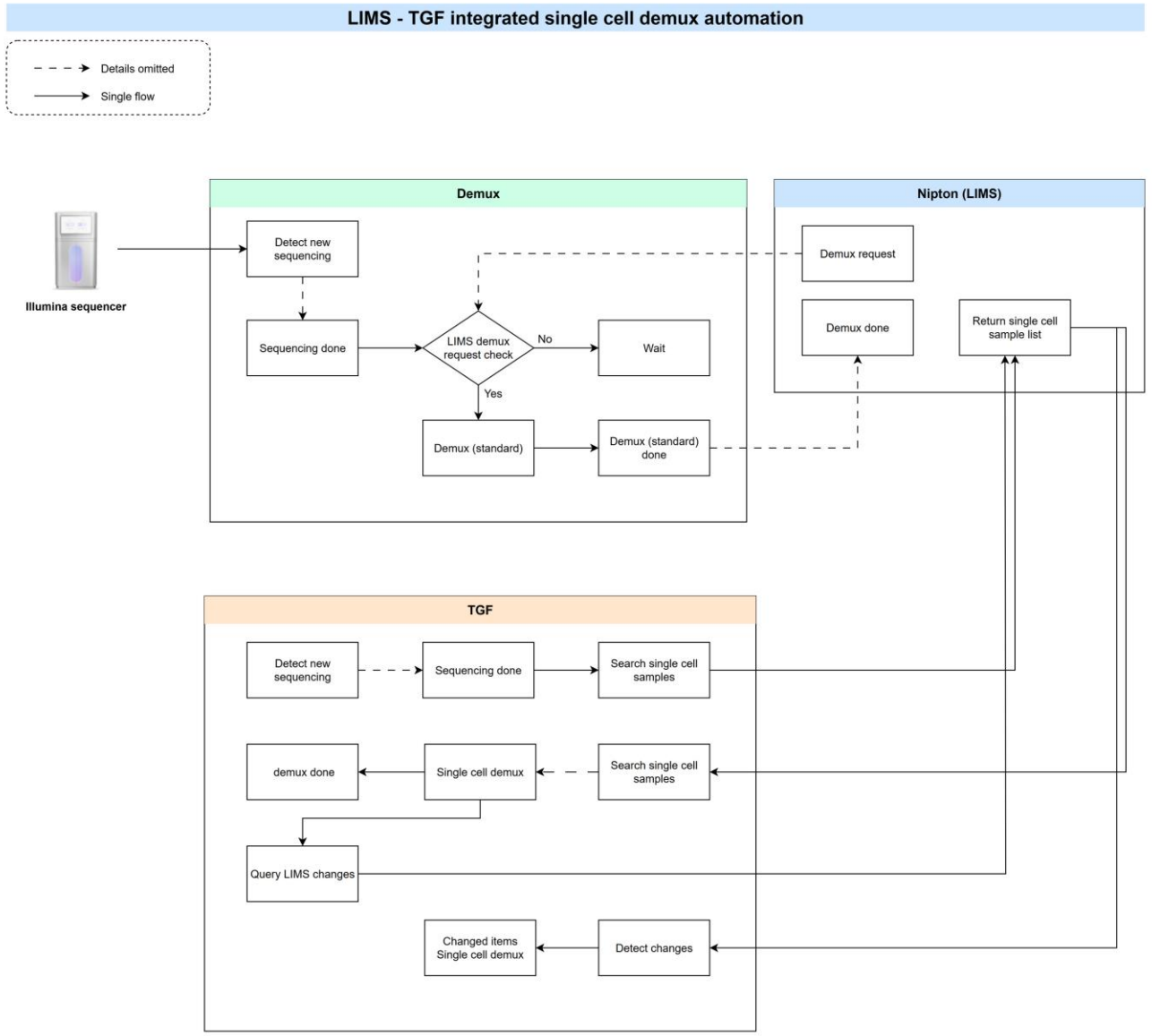
3. (Personal Project) AWS-based Audio Transcription / Translation Service

Analysis Automation Web Service - Major Workflows (TGF)

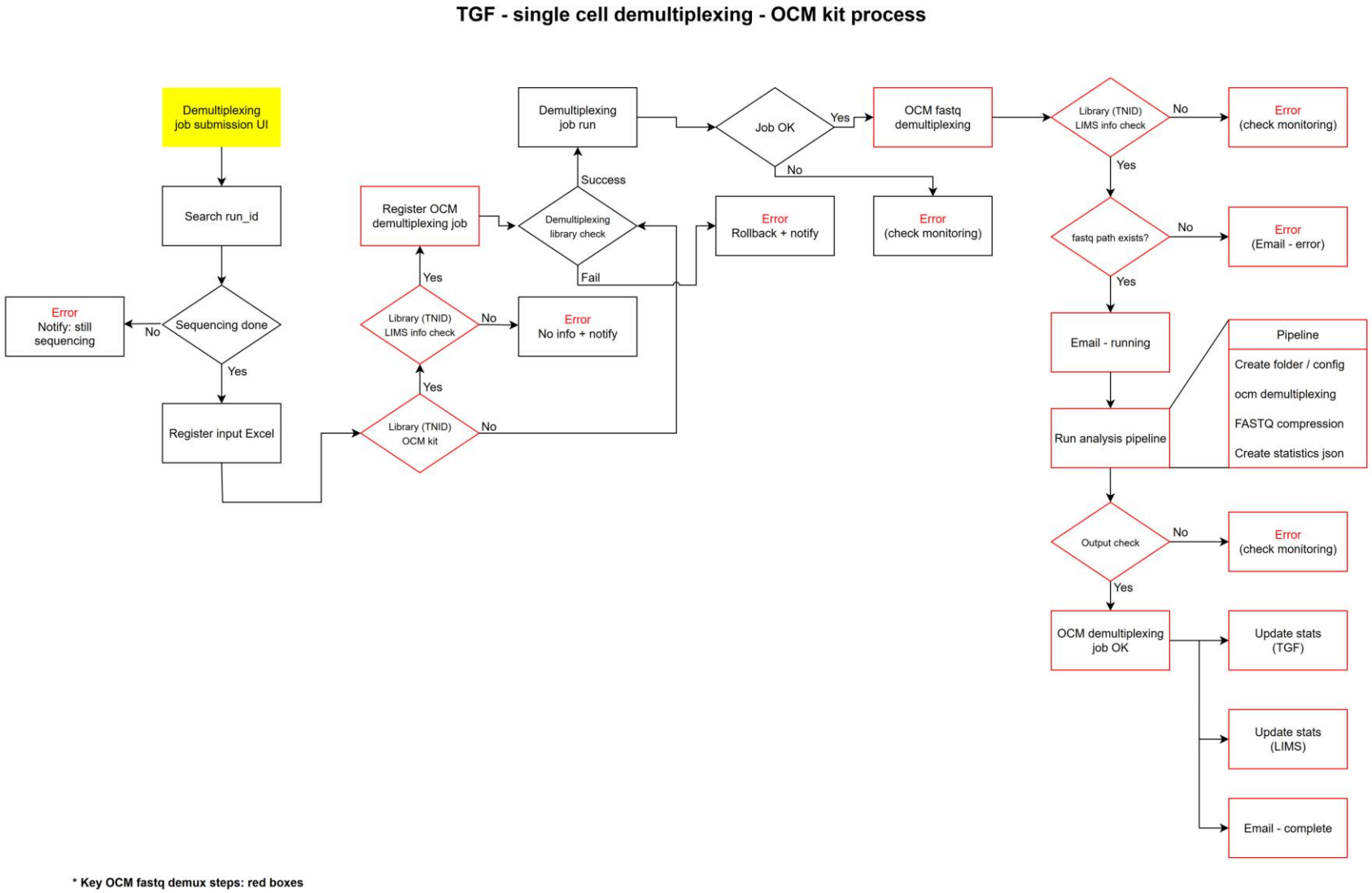
Analysis job server load control workflow



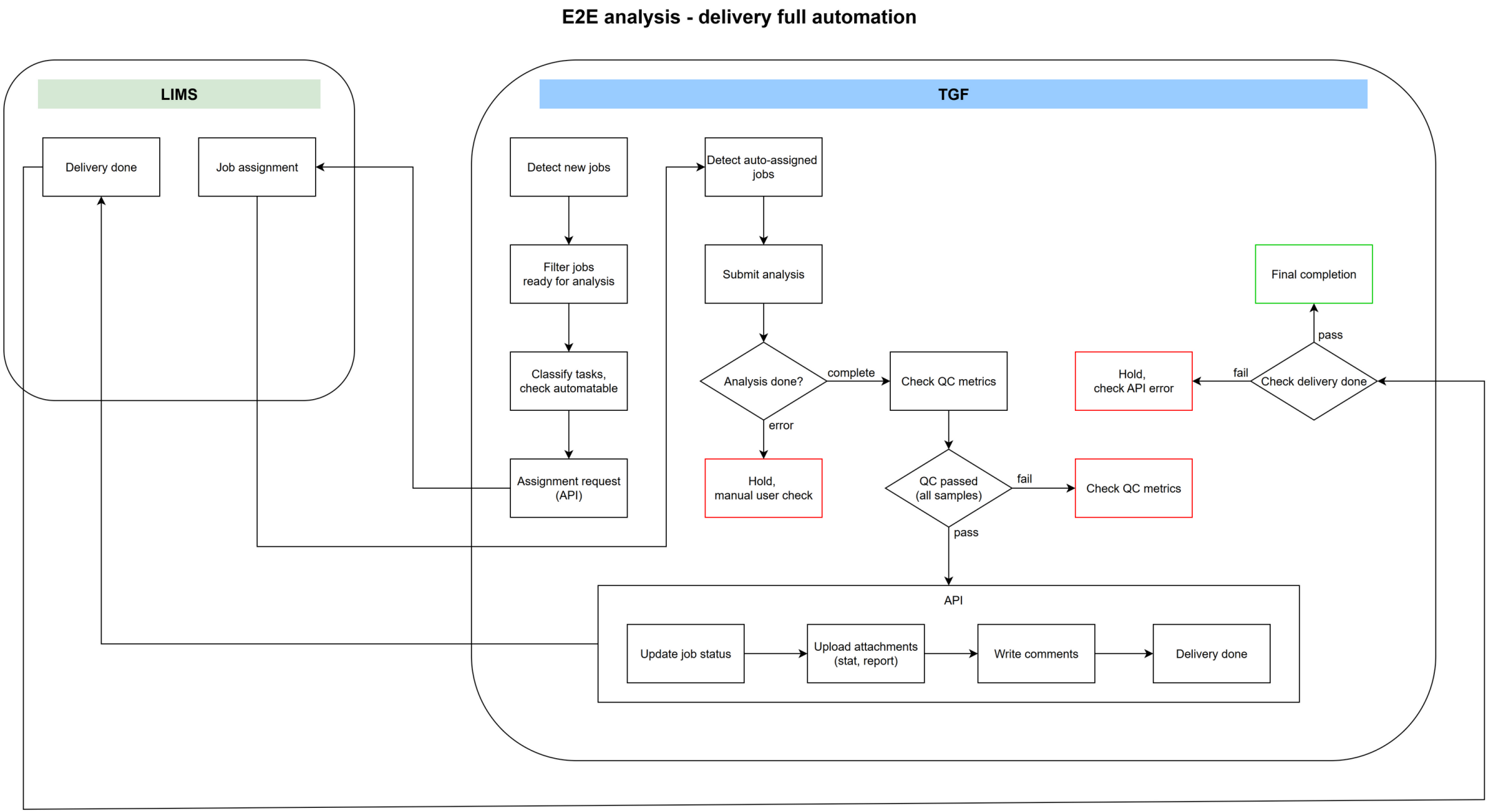
Single cell Demultiplexing Automation Workflow



Single cell - OCM fastq Demultiplexing Automation Workflow

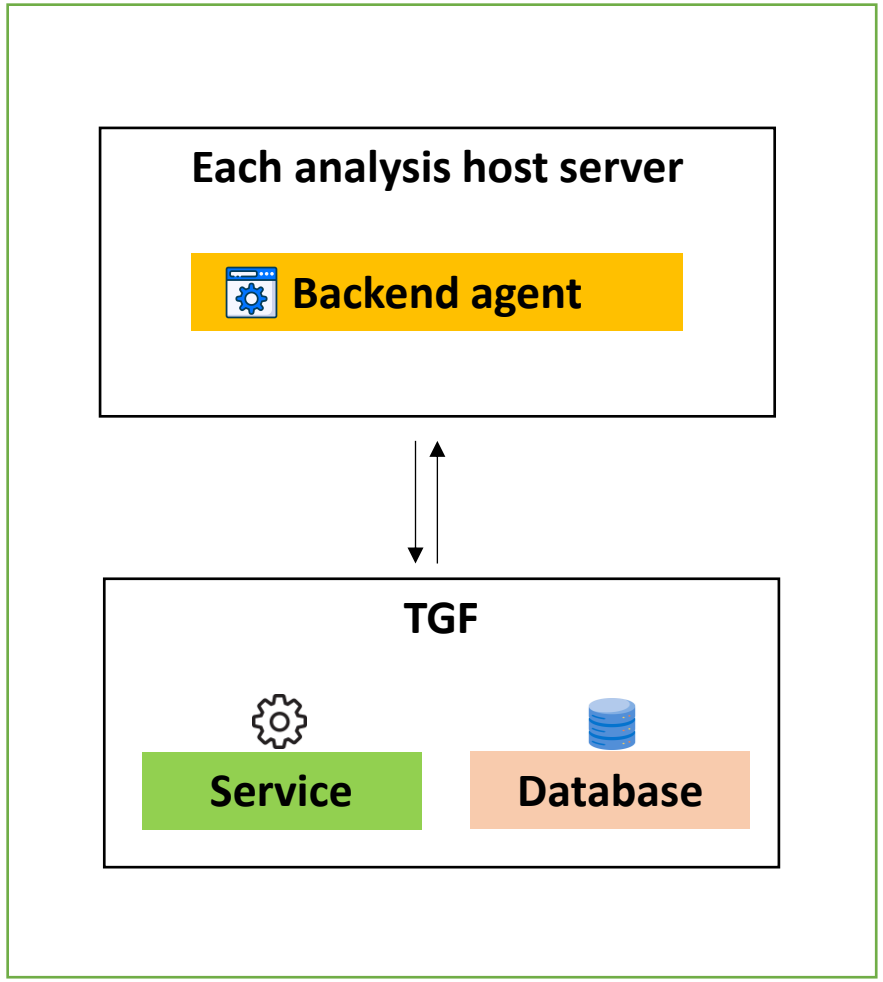


E2E analysis - delivery automation workflow



Analysis Automation Web Service - Architecture (TGF)

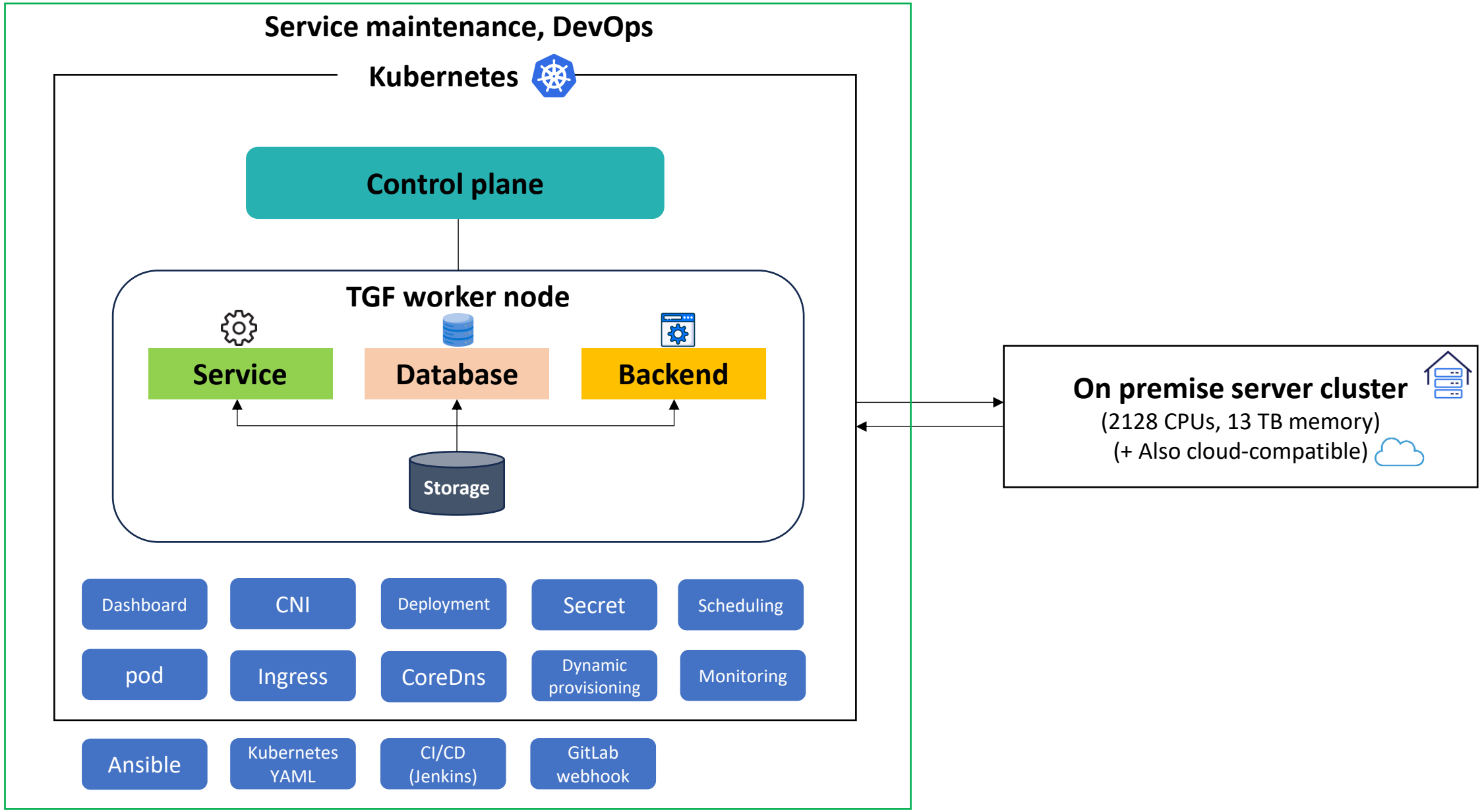
Challenges in TGF Service architecture v1



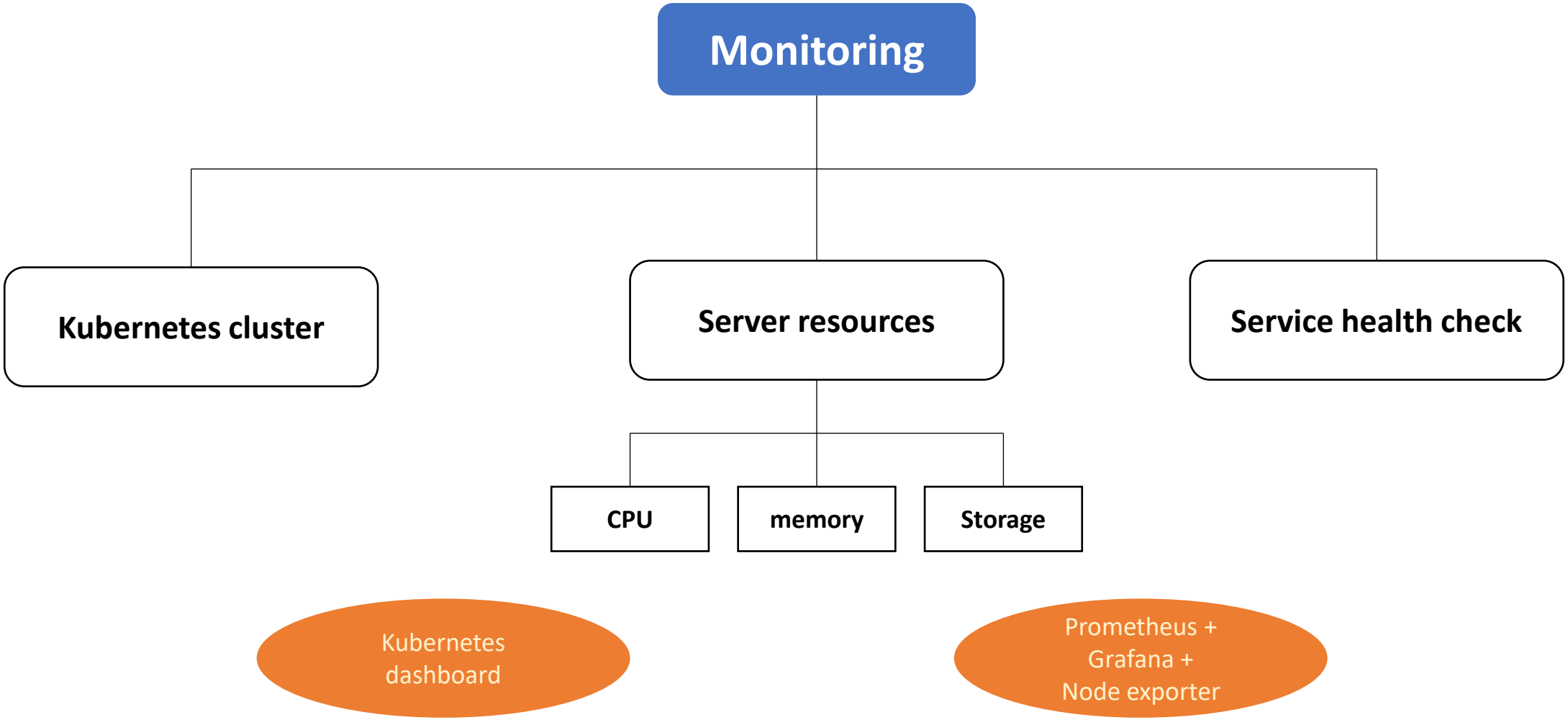
1. Code exposure
2. Security risks
3. Problems from shared server resources
   - Errors
   - Unstable service operation
   - Zombie processes
4. Agent setup needed per new host
   -> Increased maintenance

Separate servers + unified operation

TGF Service architecture v2



TGF Service architecture



Kubernetes dashboard

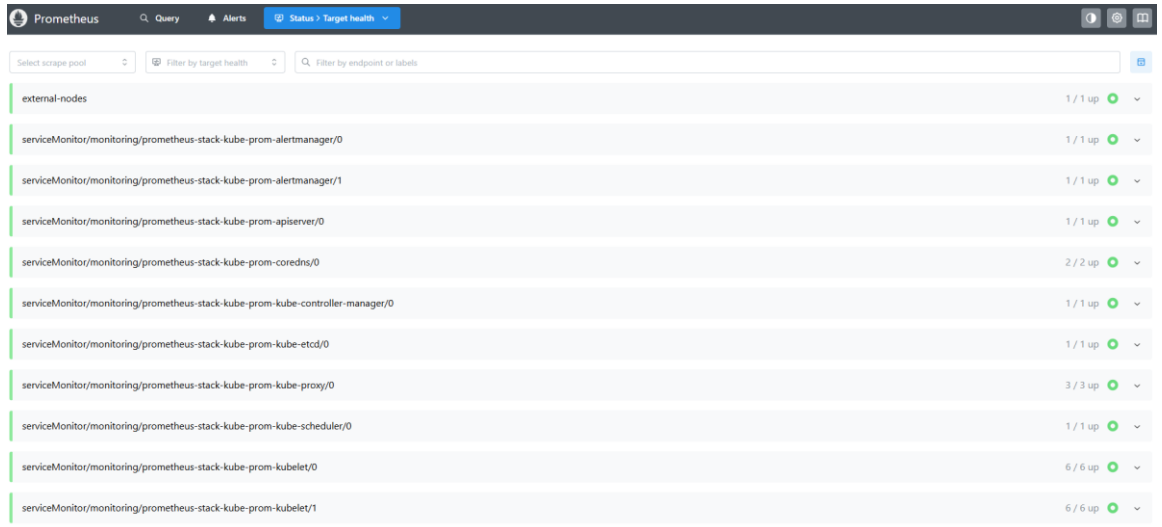


Grafana + Prometheus: Server resource monitoring

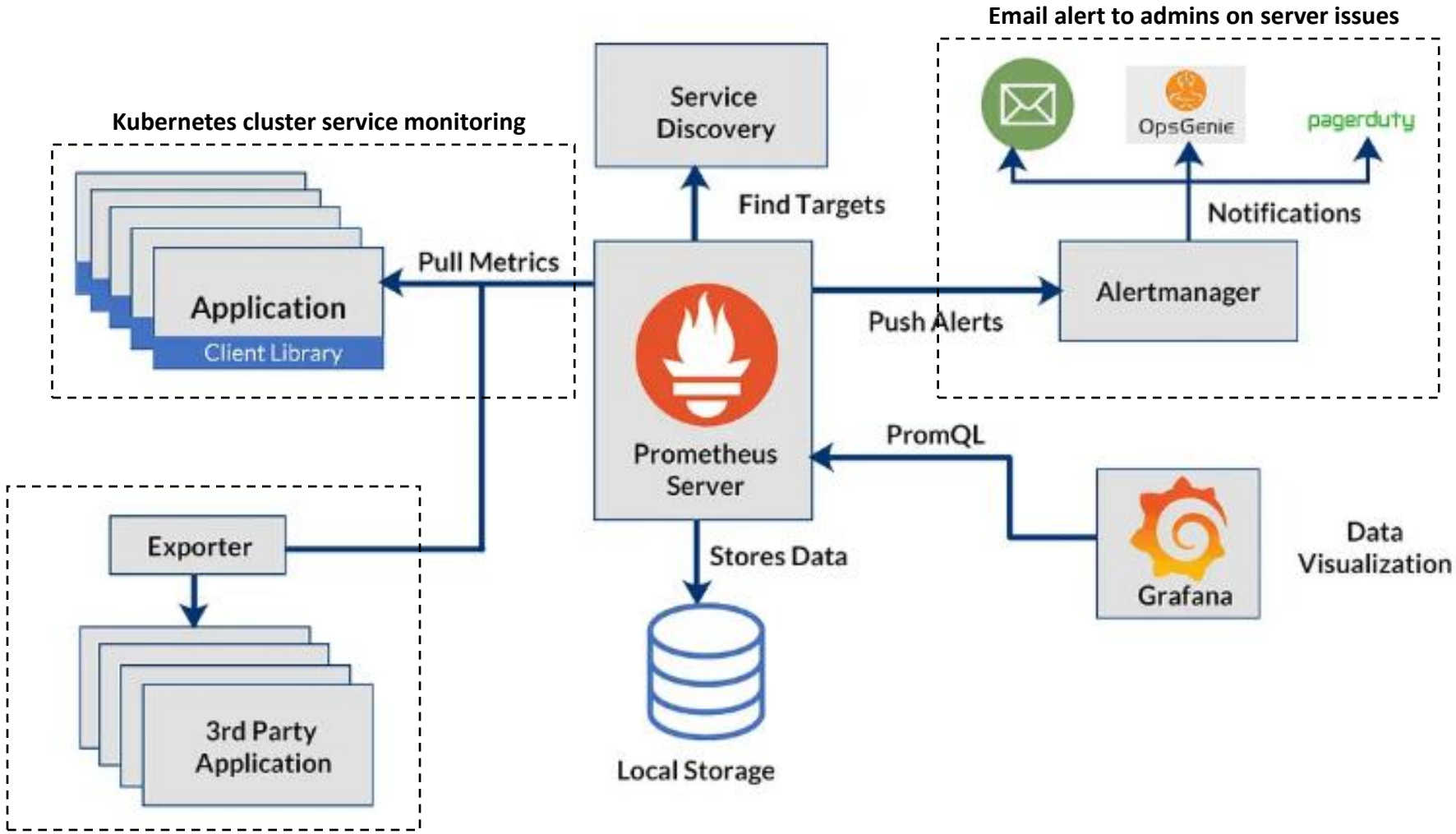
Grafana



Prometheus



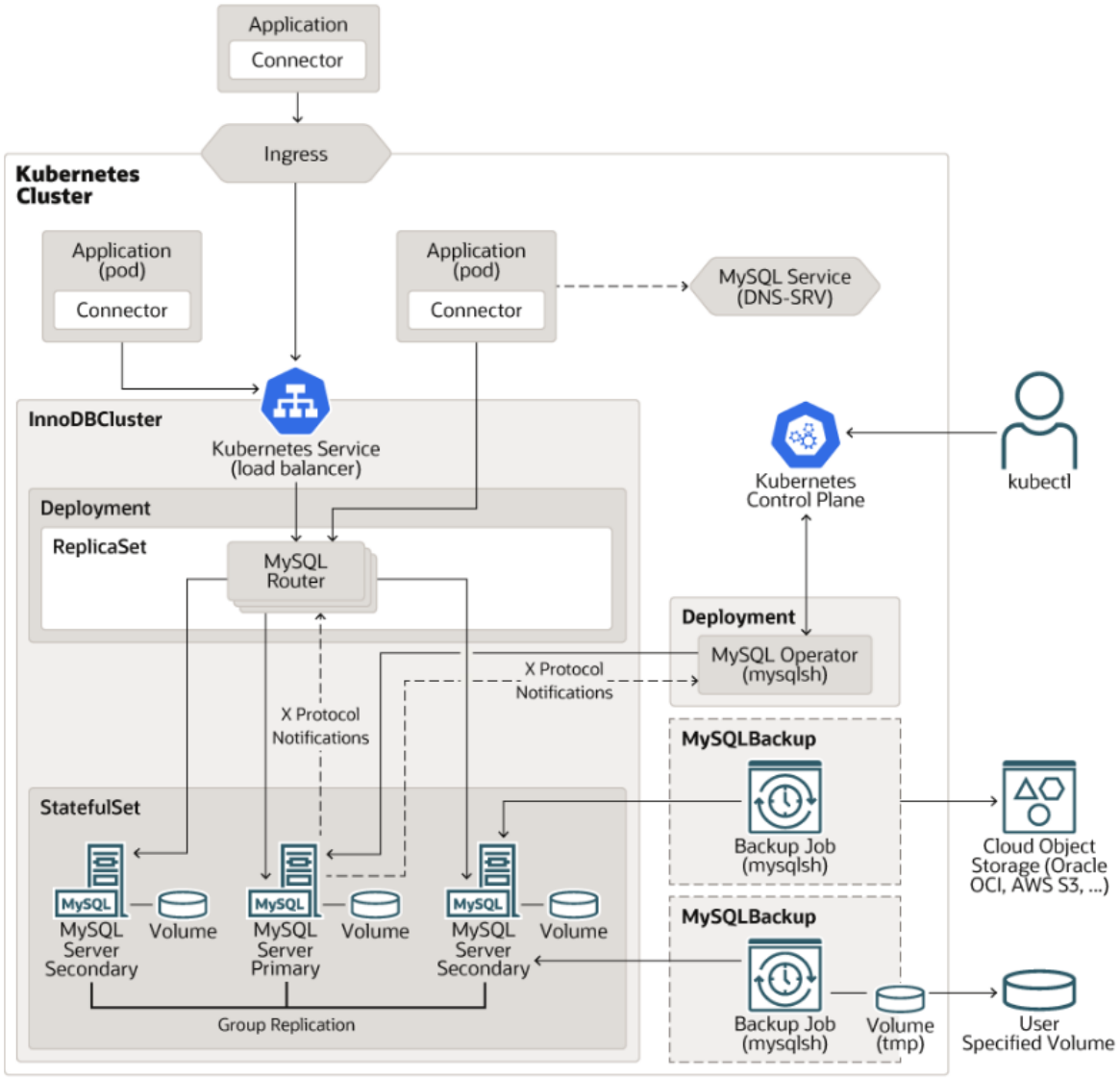
Server resource monitoring architecture



Server cluster monitoring, analysis job control based on resource usage

Database architecture

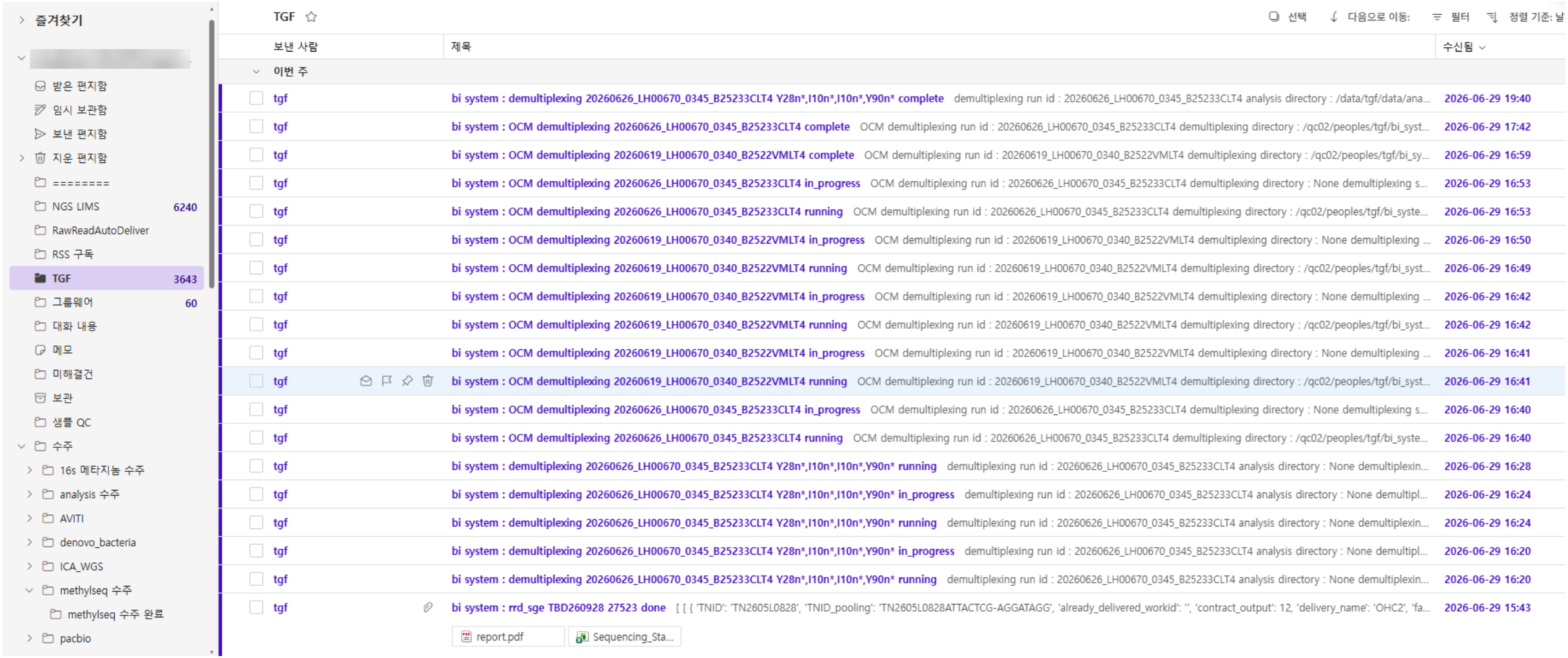
MySQL Operator for Kubernetes Architecture



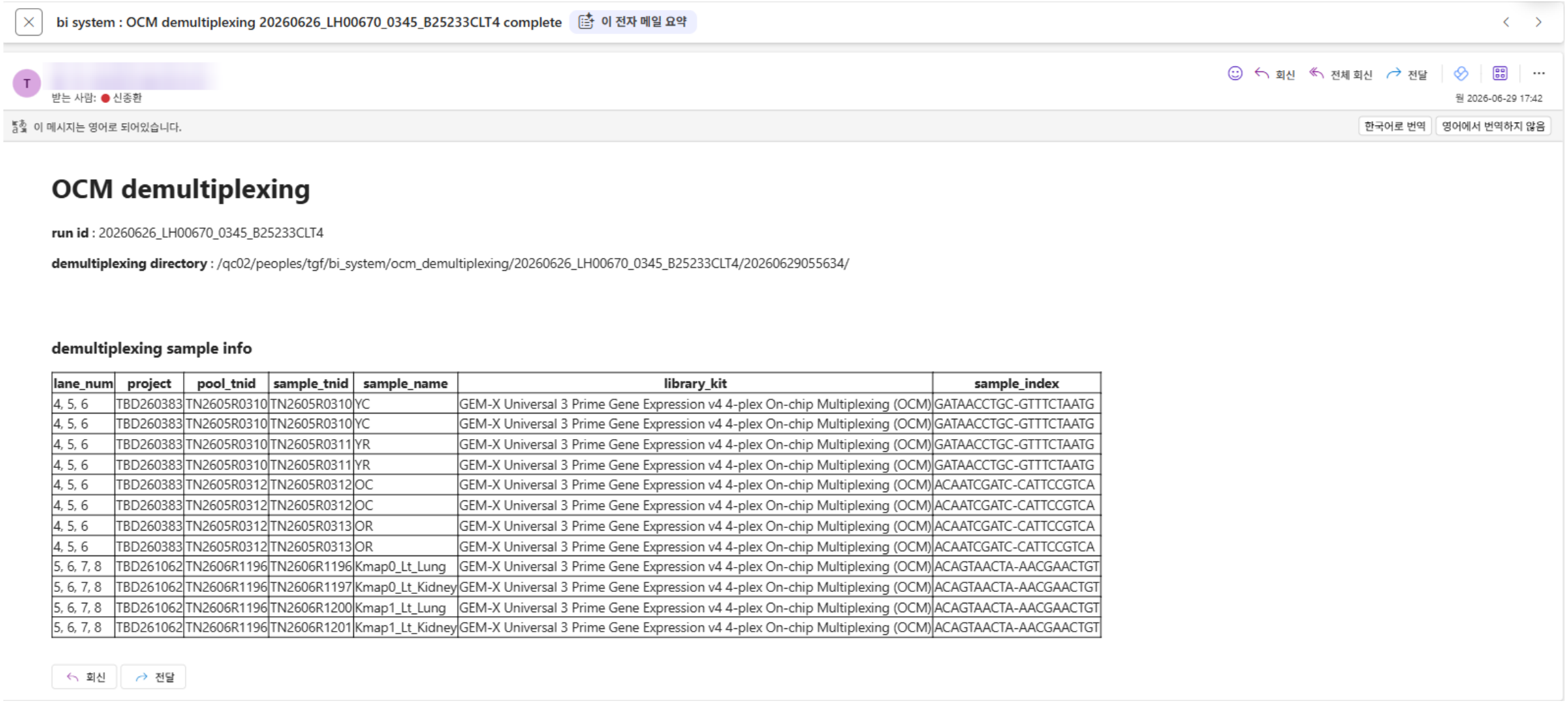
- Automated MySQL InnoDB Cluster management
- Full integration with Kubernetes
- Automated MySQL Server state recovery
- Automatic MySQL Router setup and scaling
- Clear service separation
- Customizable my.cnf, auto-applied on change
- State-based control for operational stability
- Flexible cluster scale-out / scale-in
- Declarative management (no direct Pod edits)

Email System

Each automation's process info + progress status is shown in the email subject



Email System - example (OCM demultiplexing)



bi system : OCM demultiplexing 20260626_LH00670_0345_B25233CLT4 complete

이 전자 메일 요약

회신 전체 회신 전달

받는 사람: 신종환

월 2026-06-29 17:42

이 메시지는 영어로 되어있습니다.

한국어로 번역 영어에서 번역하지 않음

OCM demultiplexing

run id : 20260626_LH00670_0345_B25233CLT4

demultiplexing directory : /qc02/peoples/tgf/bi_system/ocm_demultiplexing/20260626_LH00670_0345_B25233CLT4/20260629055634/

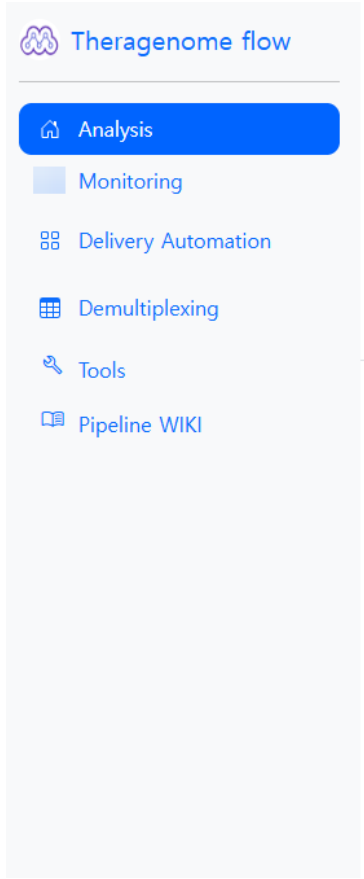
demultiplexing sample info

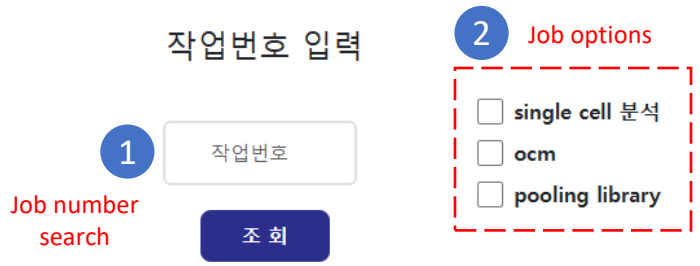
lane_num	project	pool_tnid	sample_tnid	sample_name	library_kit	sample_index
4, 5, 6	TBD260383	TN2605R0310	TN2605R0310	YC	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	GATAACCTGC-GTTTCTAATG
4, 5, 6	TBD260383	TN2605R0310	TN2605R0310	YC	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	GATAACCTGC-GTTTCTAATG
4, 5, 6	TBD260383	TN2605R0310	TN2605R0311	YR	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	GATAACCTGC-GTTTCTAATG
4, 5, 6	TBD260383	TN2605R0310	TN2605R0311	YR	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	GATAACCTGC-GTTTCTAATG
4, 5, 6	TBD260383	TN2605R0312	TN2605R0312	OC	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	ACAATCGATC-CATTCCGTCA
4, 5, 6	TBD260383	TN2605R0312	TN2605R0312	OC	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	ACAATCGATC-CATTCCGTCA
4, 5, 6	TBD260383	TN2605R0312	TN2605R0313	OR	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	ACAATCGATC-CATTCCGTCA
4, 5, 6	TBD260383	TN2605R0312	TN2605R0313	OR	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	ACAATCGATC-CATTCCGTCA
5, 6, 7, 8	TBD261062	TN2606R1196	TN2606R1196	Kmap0_Lt_Lung	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	ACAGTAACTA-AACGAACTGT
5, 6, 7, 8	TBD261062	TN2606R1196	TN2606R1197	Kmap0_Lt_Kidney	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	ACAGTAACTA-AACGAACTGT
5, 6, 7, 8	TBD261062	TN2606R1196	TN2606R1200	Kmap1_Lt_Lung	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	ACAGTAACTA-AACGAACTGT
5, 6, 7, 8	TBD261062	TN2606R1196	TN2606R1201	Kmap1_Lt_Kidney	GEM-X Universal 3 Prime Gene Expression v4 4-plex On-chip Multiplexing (OCM)	ACAGTAACTA-AACGAACTGT

회신 전달

Analysis Automation Web Service - Frontend (TGF)

Analysis





분석 샘플

번호	workid	TNID	sample_name	fastq_1	fastq_2	totalbase
1	27648	TN2606L2152	FLAG_IP_1	/rawread01/20260626_LH00670_0345_B25233CLT4/Unaligned/TBD261167/TN2606L2152--ATTCAGAA-CTTCGCCT_S40_L001_R1_001.fastq.gz	/rawread01/20260626_LH00670_0345_B25233CLT4/Unaligned/TBD261167/TN2606L2152--ATTCAGAA-CTTCGCCT_S40_L001_R2_001.fastq.gz	6,549,004,994 (6.55 Gb)
2	27648	TN2606L2153	FLAG_IP_2	/rawread01/20260626_LH00670_0345_B25233CLT4/Unaligned/TBD261167/TN2606L2153--GAATTCGT-TAAGATTA_S41_L001_R1_001.fastq.gz	/rawread01/20260626_LH00670_0345_B25233CLT4/Unaligned/TBD261167/TN2606L2153--GAATTCGT-TAAGATTA_S41_L001_R2_001.fastq.gz	7,299,416,104 (7.3 Gb)
3	27648	TN2606L2154	FLAG_IP_3	/rawread01/20260626_LH00670_0345_B25233CLT4/Unaligned/TBD261167/TN2606L2154--CTGAAGCT-ACGTCCTG_S42_L001_R1_001.fastq.gz	/rawread01/20260626_LH00670_0345_B25233CLT4/Unaligned/TBD261167/TN2606L2154--CTGAAGCT-ACGTCCTG_S42_L001_R2_001.fastq.gz	4,644,862,982 (4.64 Gb)
4	27648	TN2606L2155	FLAG_IP_4	/rawread01/20260626_LH00670_0345_B25233CLT4/Unaligned/TBD261167/TN2606L2155--TAATGCGC-GTCAGTAC_S43_L001_R1_001.fastq.gz	/rawread01/20260626_LH00670_0345_B25233CLT4/Unaligned/TBD261167/TN2606L2155--TAATGCGC-GTCAGTAC_S43_L001_R2_001.fastq.gz	4,277,305,728 (4.28 Gb)

Analysis

3 Project info

프로젝트

프로젝트	분석예정자	서비스	서비스 상세	종	매니저	의뢰기관	고객	작업번호 샘플 수	프로젝트 총 샘플 수	요청된 Gb	요청된 read 길이
TBD261167	신종환	Sequencing	Raw Data	Human	[redacted]	[redacted]	[redacted]	4	12	None	None

LIMS 참고자료(첨부파일)

4 Attachment check (preview)

1. 라이브러리 준비 방법

1-1. 농도 및 볼륨 기준

농도 (nM)	요청생산량 (Gb)	Vol. (uL)
≥ 10	~200	15 ~ 20
	200 ~ 300	20 ~ 25
	≥ 300	≥ 25

*농도가 10nM 이하일 경우, TapeStation에서 Main peak 확인이 어려움

- 과생산 또는 추가 생산 발생 확률 증가 및 data quality 보장 어려움

*요청 생산량이 많을수록 라이브러리 input 증가되어 부족할 경우, 추가 요청 할 가능성 있음

2. 의뢰서 작성 가이드

! Index 입력 주의사항 ! ■ NovaSeq 시퀀싱 플랫폼은 1개의 Lane에 최대 96개의 라이브러리를 Pooling하여 시퀀싱 진행합니다. ■ Index 정보를 잘못 전달주실 경우, 데이터가 섞이게 되어 사용이 불가능하며, ◦이로 인해 재생산이 필요하고, Data 전달 일정이 지연될 수

2-1. 소속기관, 연구대표자, 담당자, 연락처

각 항목에 맞게 텍스트로 작성한다.

*샘플 관련 문의 발생 시 필요한 정보이므로 정확하게 입력 바랍니다.

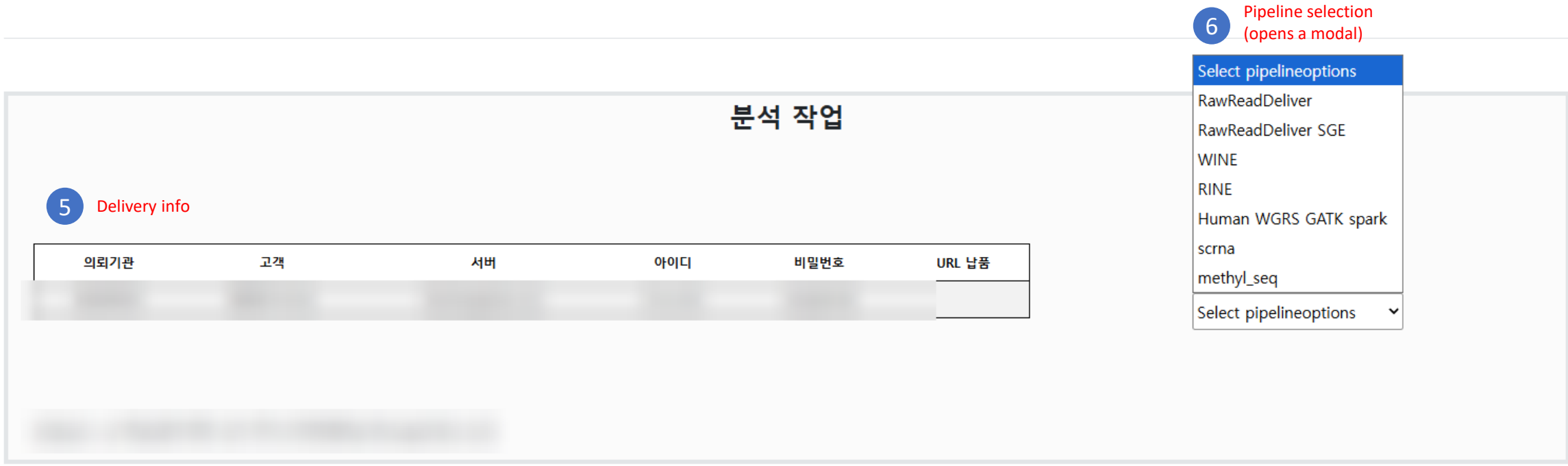
2-2. Sample Species

작성 가이드라인 | Library Running Info. Templ | 드롭다운 메뉴

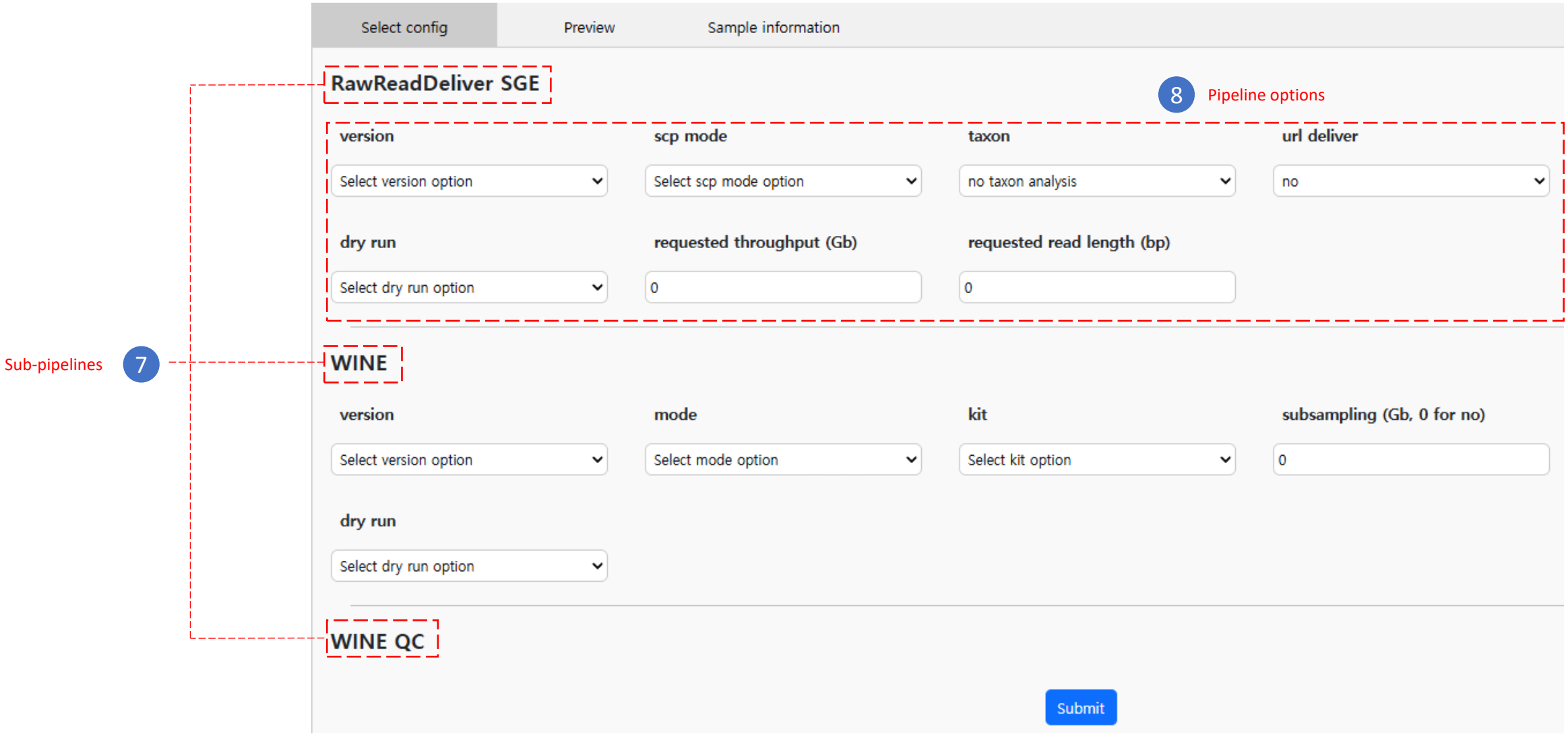
서비스 로그

없음

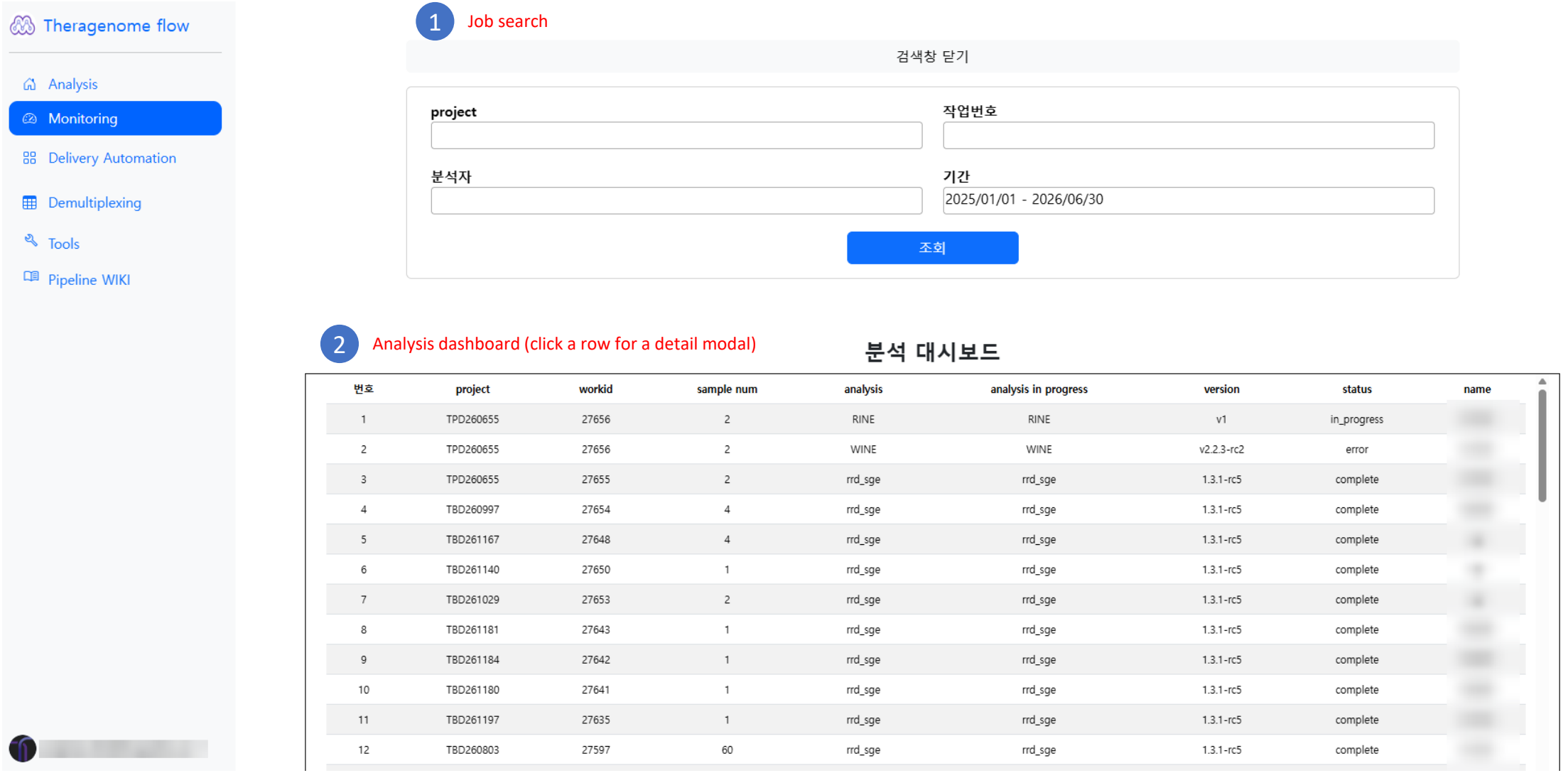
Analysis


6
Pipeline selection
(opens a modal)
Select pipelineoptions
RawReadDeliver
RawReadDeliver SGE
WINE
RINE
Human WGRS GATK spark
scrna
methyl_seq
Select pipelineoptions
분석 작업
5
Delivery info
의뢰기관
고객
서버
아이디
비밀번호
URL 납품

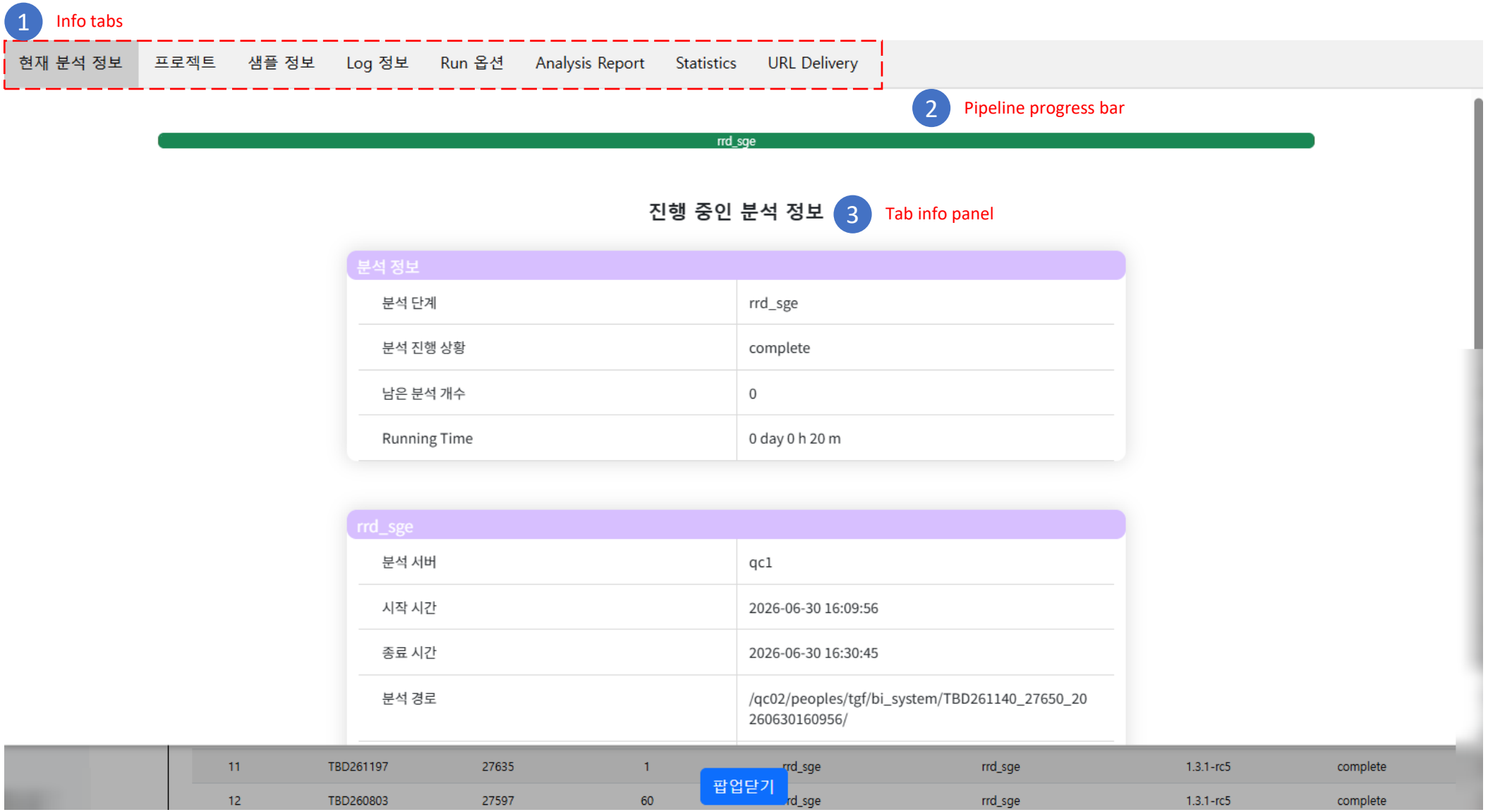
Analysis - Pipeline Selection



Analysis Monitoring Dashboard



Analysis Monitoring Dashboard - Analysis Details



Analysis Monitoring Dashboard - Analysis Details (log)

현재 분석 정보 | 프로젝트 | 샘플 정보 | Log 정보 | Run 옵션 | Analysis Report | Statistics | URL Delivery

Log 정보

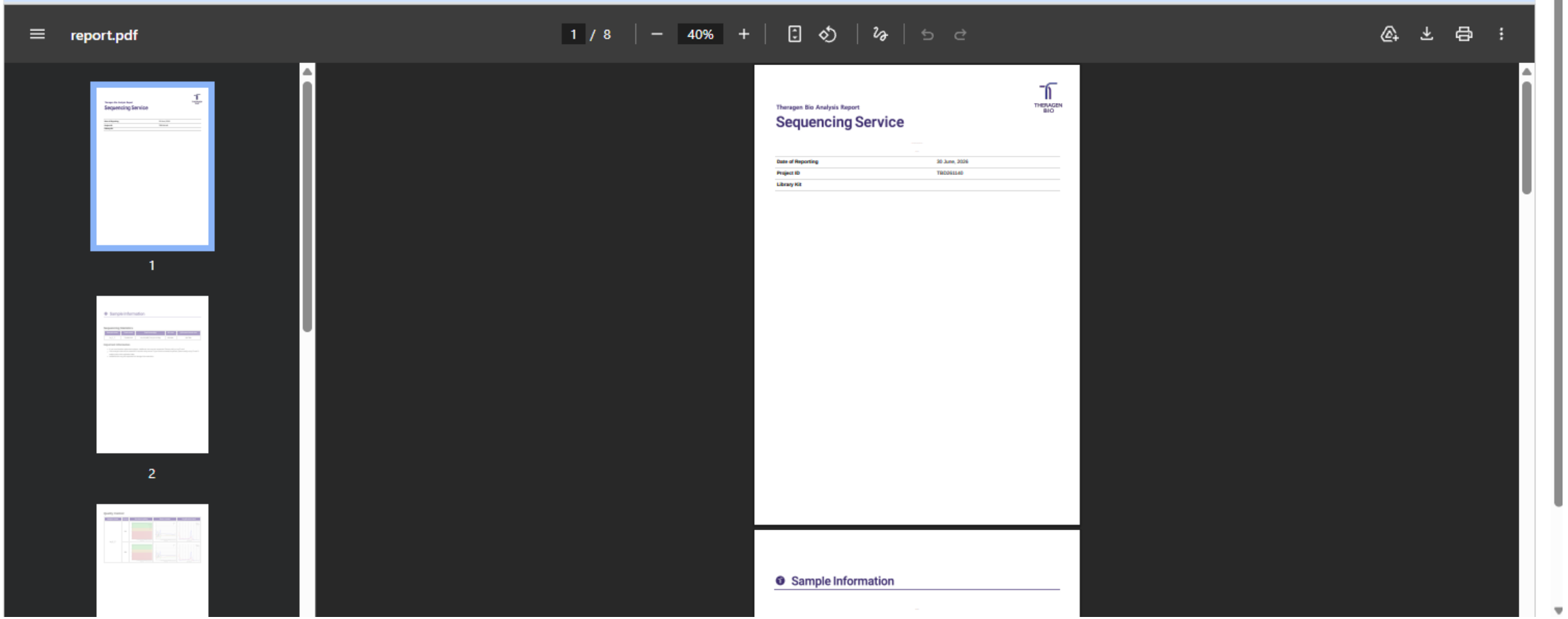
rrd_sge 4 Real-time log for each pipeline (no delay)

args.mode : db_update
sDb_update_mode in_progress
[redacted]
dicData_check_config {'single': {'report_pdf': {'idx': 33, 'pipeline_idx_fk': 8, 'pipeline': 'rrd_sge', 'check_type': 'single', 'output_name': 'report_pdf', 'output_path': 'results/report/report.pdf', 'output_type': 'report', 'output_type_dest_tb': 'analysis_log', 'output_type_dest_col': 'report_file', 'output_file_type': None, 'output_metadata_id': None, 'output_start_line': None, 'transfer_yn': 1, 'email_attach_yn': 1},
[redacted]
'output_type_dest_tb': None, 'output_type_dest_col': None, 'output_file_type': None, 'output_metadata_id': None, 'output_start_line': None, 'transfer_yn': 1, 'email_attach_yn': 1}, 'seq_stat_xls': {'idx': 35, 'pipeline_idx_fk': 8, 'pipeline': 'rrd_sge', 'check_type': 'single', 'output_name': 'seq_stat_xls', 'output_path': 'results/fastqstat/Sequencing_Statistics_Result.xls', 'output_type': 'stat', 'output_type_dest_tb': None, 'output_type_dest_col': None, 'output_file_type': 'tsv', 'output_metadata_id': None, 'output_start_line': 1, 'transfer_yn': 1, 'email_attach_yn': 0}, 'seq_stat_xls_sc': {'idx': 36, 'pipeline_idx_fk': 8, 'pipeline': 'rrd_sge', 'check_type': 'single', 'output_name': 'seq_stat_xls_sc', 'output_path': 'results/fastqstat/Sequencing_Statistics_Result.xls', 'output_type': 'stat', 'output_type_dest_tb': None, 'output_type_dest_col': None, 'output_file_type': 'tsv', 'output_metadata_id': None, 'output_start_line': 1, 'transfer_yn': 1, 'email_attach_yn': 0}, 'taxon': {'idx': 64, 'pipeline_idx_fk': 8, 'pipeline': 'rrd_sge', 'check_type': 'single', 'output_name': 'taxon', 'output_path': 'results/taxon/TaxonomyProfiling_Statistics_Result.xls', 'output_type': 'stat', 'output_type_dest_tb': None, 'output_type_dest_col': None, 'output_file_type': 'tsv', 'output_metadata_id': None, 'output_start_line': 1, 'transfer_yn': 1, 'email_attach_yn': 0}}, 'sample': {'fastq_1': {'idx': 37, 'pipeline_idx_fk': 8, 'pipeline': 'rrd_sge', 'check_type': 'sample', 'output_name': 'fastq_1', 'output_path': 'results/ftp_upload/Sample_[sample_name]/[sample_name]_1.fq.gz', 'output_type': None, 'output_type_dest_tb': None, 'output_type_dest_col': None, 'output_file_type': None, 'output_metadata_id': None, 'output_start_line': None, 'transfer_yn': None, 'email_attach_yn': 0}, 'fastq_1_pooling': {'idx': 41, 'pipeline_idx_fk': 8, 'pipeline': 'rrd_sge', 'check_type': 'sample', 'output_name': 'fastq_1_pooling', 'output_path': "results/ftp_upload/Sample_[library_index_alias_use.pycheck_pooling_library('{self}')]/[library_index_alias_use.pycheck_pooling_library('{self}')]_1.fq.gz", 'output_type': None, 'output_type_dest_tb': None, 'output_type_dest_col': None, 'output_file_type': None, 'output_metadata_id': None, 'output_start_line': None, 'transfer_yn': None, 'email_attach_yn': 0}, 'fastq_1_sc': {'idx': 46, 'pipeline_idx_fk': 8, 'pipeline': 'rrd_sge', 'check_type': 'sample', 'output_name': 'fastq_1_sc', 'output_path': "results/ftp_upload/Sample_[sample_name]/[sample_name]_S1_L001_R1_001.fastq.gz[sc_fastq_r1.pyraise_error_string_check('{self}', 'none')]", 'output_type': None, 'output_type_dest_tb': None, 'output_type_dest_col': None, 'output_file_type': None, 'output_metadata_id': None, 'output_start_line': None, 'transfer_yn': None, 'email_attach_yn': 0}, 'fastq_2': {'idx': 38, 'pipeline_idx_fk': 8, 'pipeline': 'rrd_sge', 'check_type': 'sample', 'output_name': 'fastq_2', 'output_path': 'results/ftp_upload/Sample_[sample_name]/[sample_name]_2.fq.gz', 'output_type': None, 'output_type_dest_tb':

Analysis Monitoring Dashboard - Analysis Details (Analysis Report)

현재 분석 정보 | 프로젝트 | 샘플 정보 | Log 정보 | Run 옵션 | Analysis Report | Statistics | URL Delivery

rrd_sge 5 Report view after each pipeline completes



Analysis Monitoring Dashboard - Analysis Details (Analysis Report)

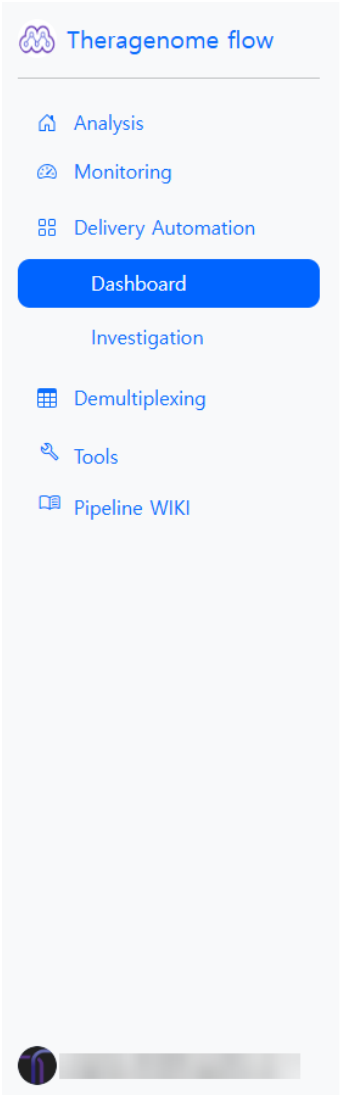
현재 분석 정보 | 프로젝트 | 샘플 정보 | Log 정보 | Run 옵션 | Analysis Report | Statistics | URL Delivery

rrd_sge

sequencing statistics 6 Stats view after each pipeline completes (by pipeline and by statistics type)

SampleID	TBI_ID	TotalReads	TotalBases	TotalBases(Gb)	GC_Count	GC_Rate	N_ZeroReads	N_ZeroReadsRate	N5_LessReads	N5_LessReadsRate	N_Count	N_Rate	Q30_MoreBases	Q30_MoreBasesRate	Q20_MoreBases	Q
N_C_T	TN2606L1346	73662210	11122993710	11.12 Gb	6691873648	60.16%	72835224	98.88%	73660822	100.00%	534242	0.00%	10542947232	94.79%	10943958379	

Delivery Automation (E2E analysis-to-delivery automation)



데이터 납품 자동화

작업번호	프로젝트	의뢰기관 / 의뢰자	분류	파이프라인	분석	등록일	상태	유저 판단	제출자	상세
27648	TBD261167	[redacted]	FTP	rrd_sge	분석 #3675 ↗	2026-06-30 14:45	진행 중	확인	tgf	상세
27650	TBD261140	[redacted]	URL	rrd_sge	분석 #3674 ↗	2026-06-30 14:45	진행 중	미확인	tgf	상세
27653	TBD261029	[redacted]	FTP	rrd_sge	분석 #3673 ↗	2026-06-30 14:45	진행 중	미확인	tgf	상세
27618	TBD261121	[redacted]	single cell FTP	rrd_sge	분석 #3669 ↗	2026-06-30 12:15	납품 가능	확인	신종환	상세
27609	TBD261036	[redacted]	single cell FTP	rrd_sge	분석 #3666 ↗	2026-06-29 10:50	납품 가능	확인	tgf	상세
27610	TBD261037	[redacted]	single cell FTP	rrd_sge	분석 #3665 ↗	2026-06-29 10:50	납품 가능	확인	tgf	상세
27612	TBD261077	[redacted]	single cell FTP	rrd_sge	분석 #3664 ↗	2026-06-29 10:49	납품 가능	확인	tgf	상세
27613	TBD260970	[redacted]	FTP	rrd_sge	분석 #3663 ↗	2026-06-29 10:49	납품 가능	확인	tgf	상세
27623	TBD261007	[redacted]	FTP	rrd_sge	분석 #3662 ↗	2026-06-29 10:49	보류	확인	tgf	상세
27624	TBD261144	[redacted]	FTP	rrd_sge	분석 #3661 ↗	2026-06-29 10:49	납품 가능	확인	tgf	상세
27625	TBD261148	[redacted]	FTP	rrd_sge	분석 #3660 ↗	2026-06-29 10:49	납품 가능	확인	tgf	상세
27628	TBD261163	[redacted]	single cell FTP	rrd_sge	분석 #3659 ↗	2026-06-29 10:49	납품 가능	확인	tgf	상세
27604	TBD261002	[redacted]	pooling FTP	rrd_sge	분석 #3658 ↗	2026-06-26 15:24	납품 가능	미확인	tgf	상세
27548	TBD260777	[redacted]	FTP	rrd_sge	분석 #3655 ↗	2026-06-26 09:55	보류	미확인	tgf	상세
27523	TBD260928	[redacted]	pooling FTP	rrd_sge	분석 #3657 ↗	2026-06-26 09:35	납품 가능	미확인	tgf	상세
27587	TBD261142	[redacted]	FTP	-	-	2026-06-24 11:15	보류	미확인	tgf	상세

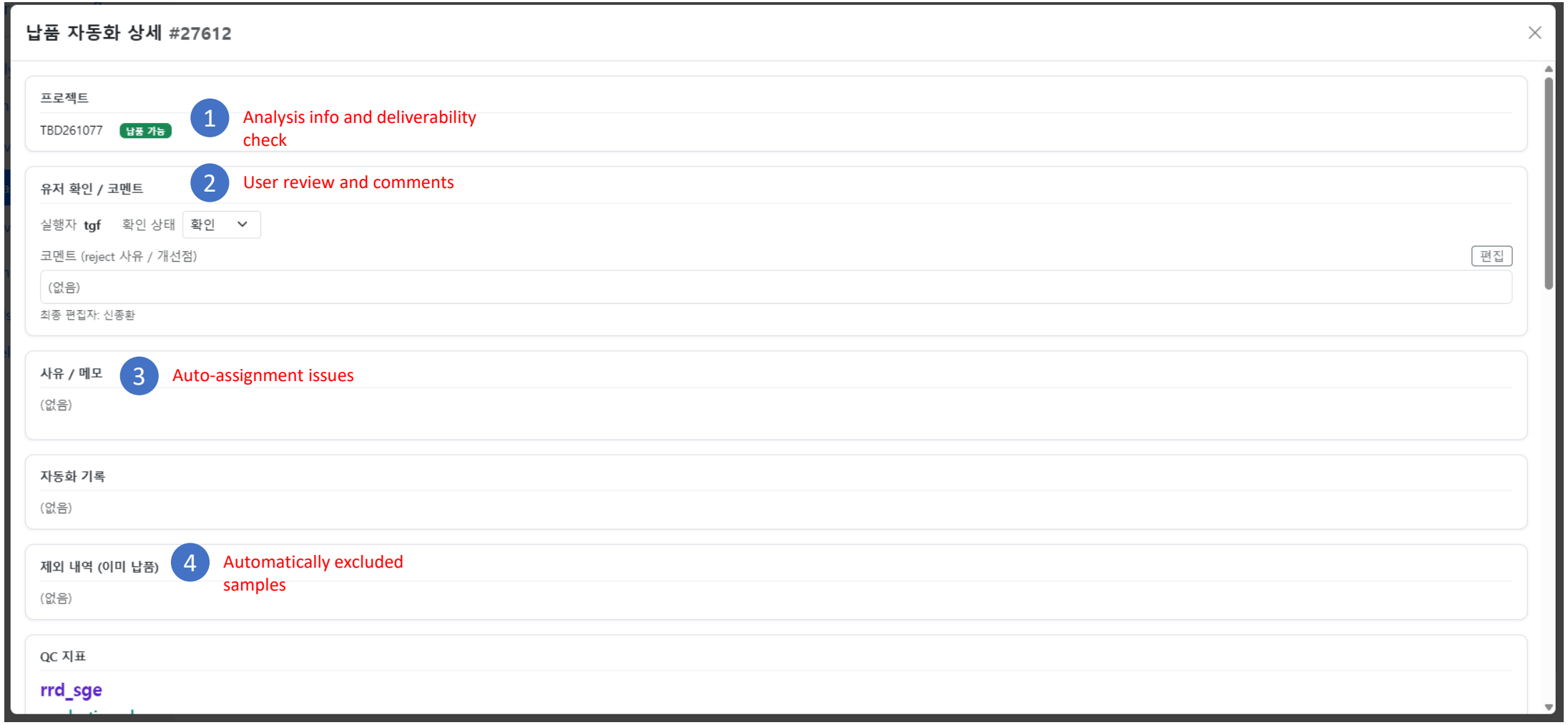
1 Analysis info

2 Automatic analysis classification

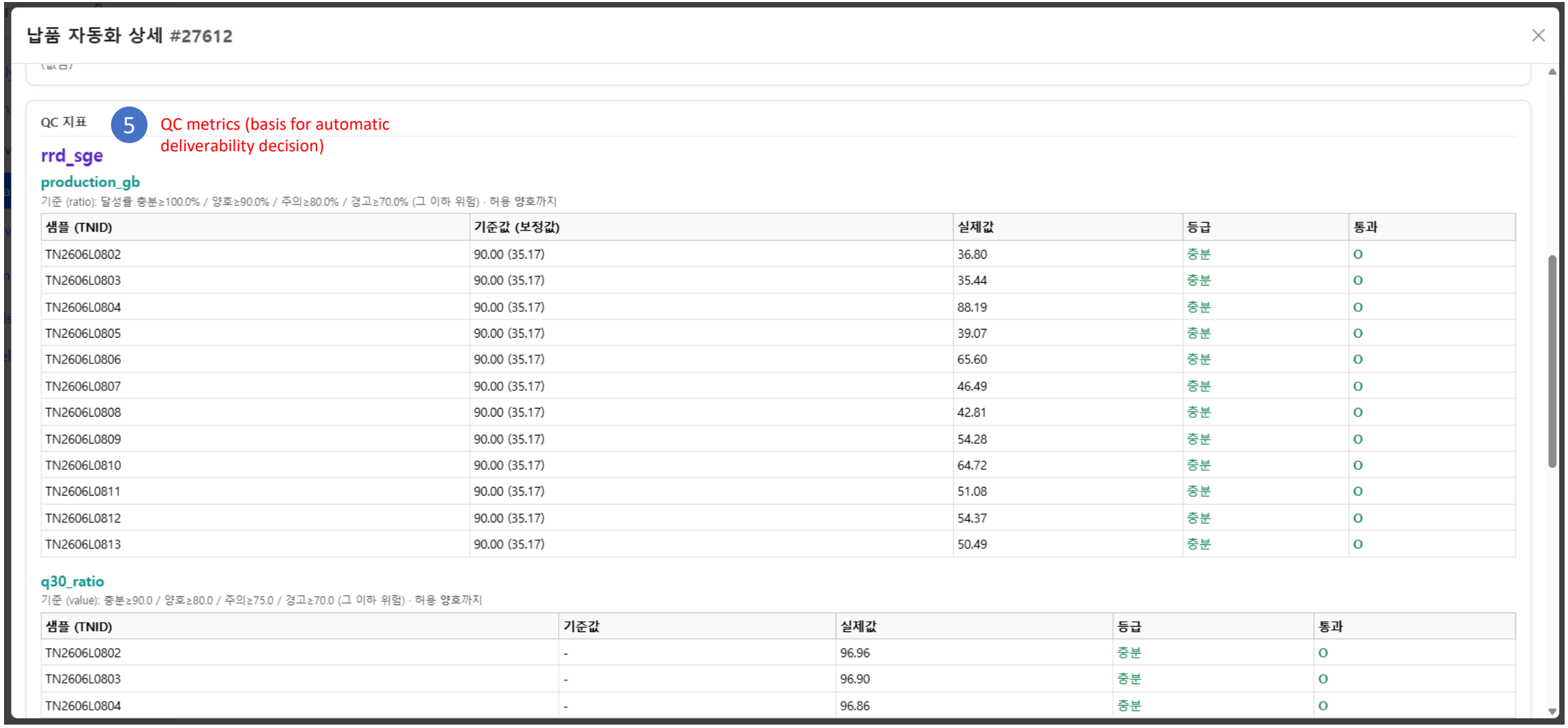
3 Analysis dashboard link

4 Automation details

Delivery Automation (E2E analysis-to-delivery automation) - Detail Modal



Delivery Automation (E2E analysis-to-delivery automation) - Detail Modal



납품 자동화 상세 #27612

QC 지표

5 QC metrics (basis for automatic deliverability decision)

rrd_sge

production_gb

기준 (ratio): 달성률 충분≥100.0% / 양호≥90.0% / 주의≥80.0% / 경고≥70.0% (그 이하 위험) · 허용 양호까지

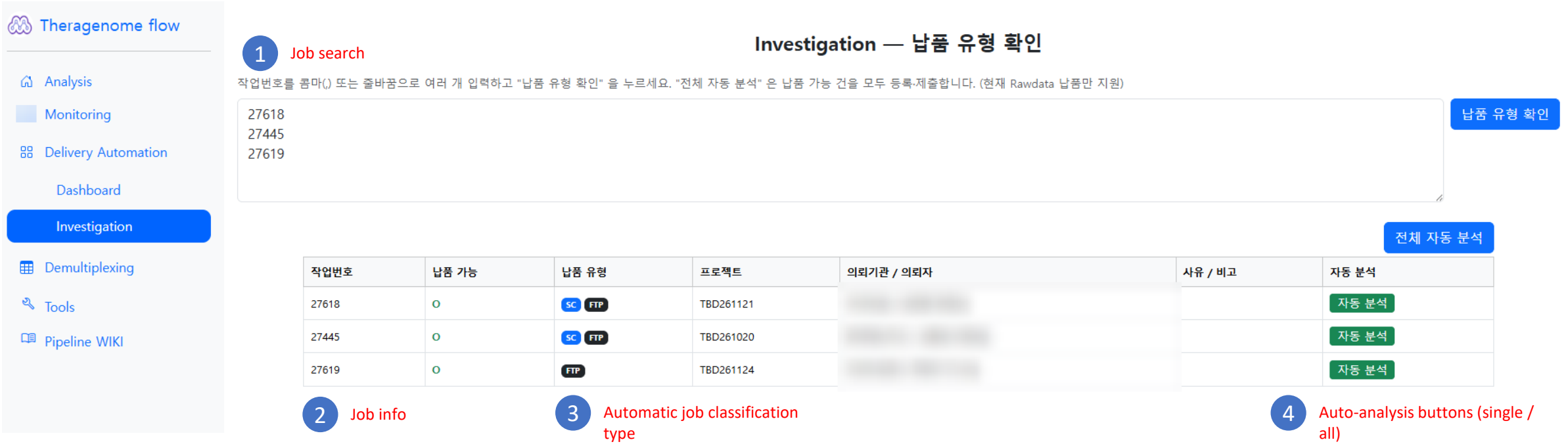
샘플 (TNID)	기준값 (보정값)	실제값	등급	통과
TN2606L0802	90.00 (35.17)	36.80	충분	O
TN2606L0803	90.00 (35.17)	35.44	충분	O
TN2606L0804	90.00 (35.17)	88.19	충분	O
TN2606L0805	90.00 (35.17)	39.07	충분	O
TN2606L0806	90.00 (35.17)	65.60	충분	O
TN2606L0807	90.00 (35.17)	46.49	충분	O
TN2606L0808	90.00 (35.17)	42.81	충분	O
TN2606L0809	90.00 (35.17)	54.28	충분	O
TN2606L0810	90.00 (35.17)	64.72	충분	O
TN2606L0811	90.00 (35.17)	51.08	충분	O
TN2606L0812	90.00 (35.17)	54.37	충분	O
TN2606L0813	90.00 (35.17)	50.49	충분	O

q30_ratio

기준 (value): 충분≥90.0 / 양호≥80.0 / 주의≥75.0 / 경고≥70.0 (그 이하 위험) · 허용 양호까지

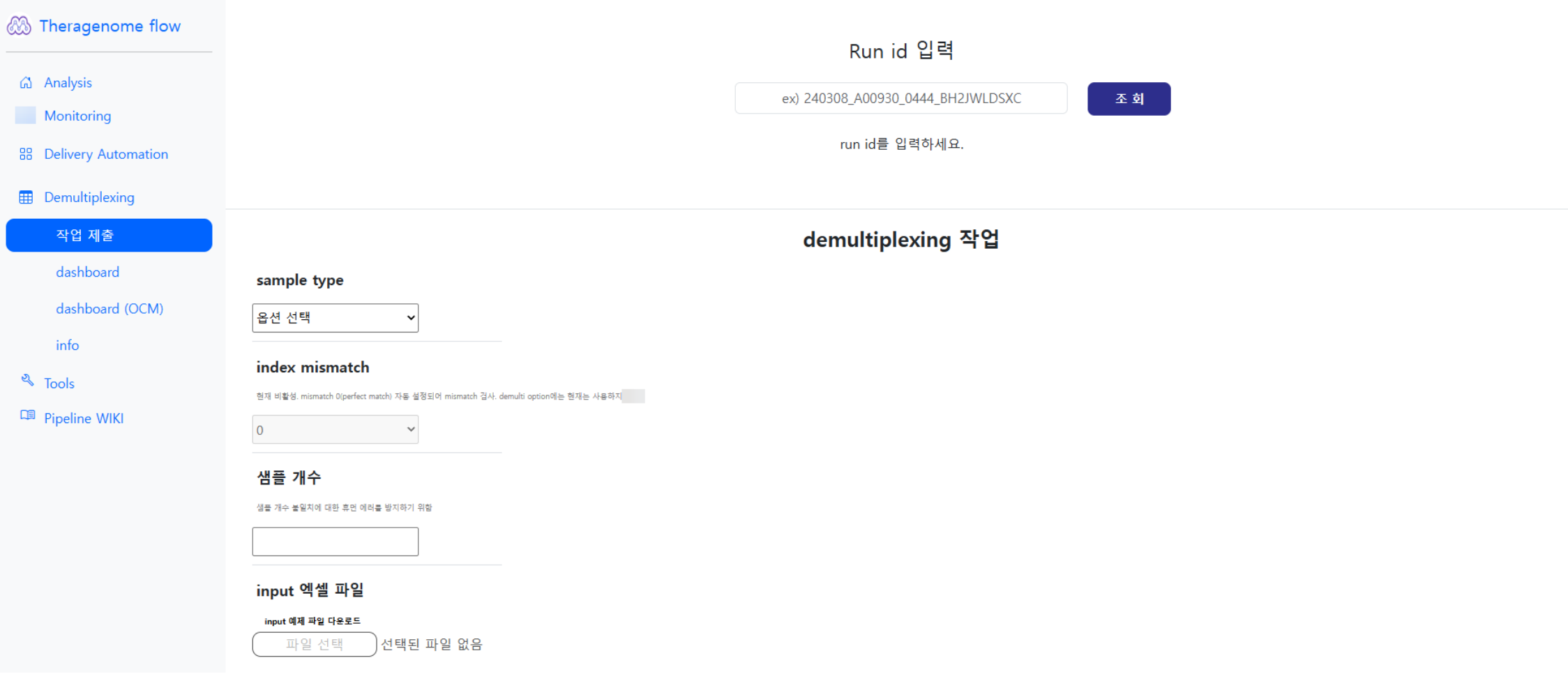
샘플 (TNID)	기준값	실제값	등급	통과
TN2606L0802	-	96.96	충분	O
TN2606L0803	-	96.90	충분	O
TN2606L0804	-	96.86	충분	O

Delivery Automation - Investigation (checks whether jobs can be auto-analyzed)

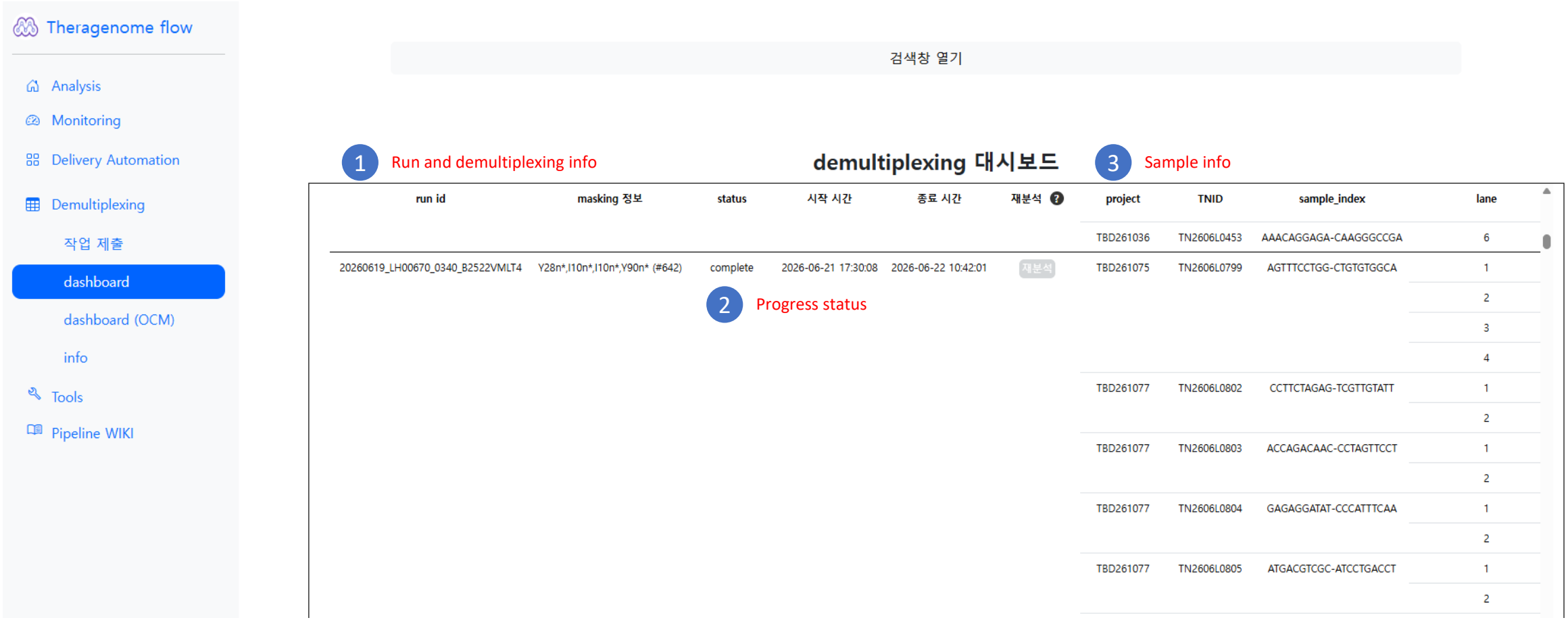


Demultiplexing - Job Submission

*** Now fully automated; used only when a manual submission is needed**



Demultiplexing - Dashboard



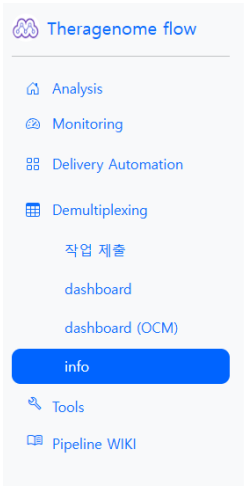
Demultiplexing - Dashboard

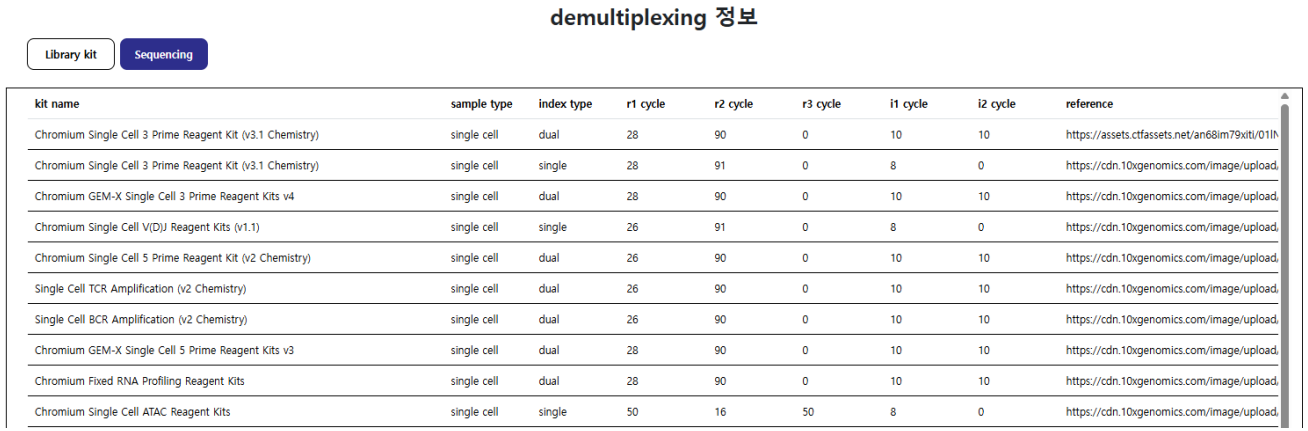
Submitter
(here: auto-submitted by the system) 5

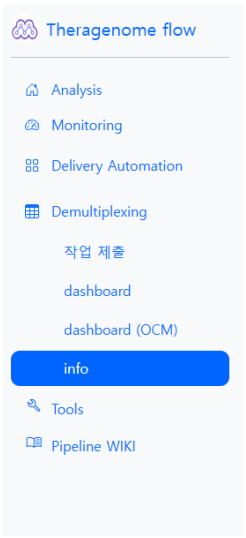
demultiplexing 대시보드

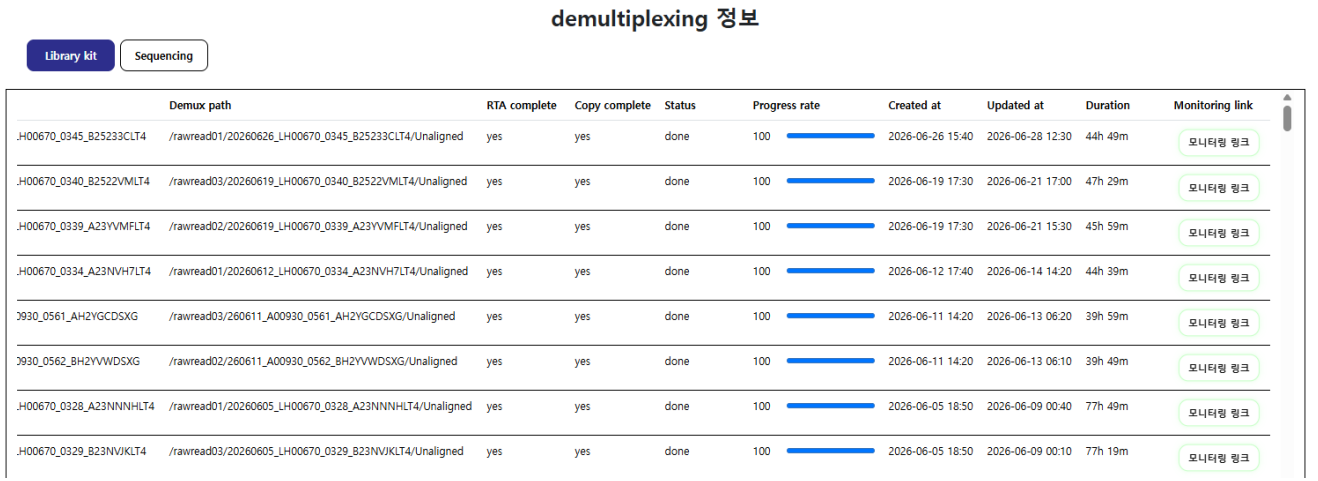
ect	TNID	sample_index	lane	yield(Gb) 4 Output yield	경로	제출자
i1075	TN2606L0799	AGTTTCCTGG-CTGTGTGGCA	1	9.86	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	AUTO_SC_SCAN
			2	11.43	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			3	12.54	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			4	15.08	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
i1077	TN2606L0802	CCTTCTAGAG-TCGTTGTATT	1	14.64	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			2	16.24	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
i1077	TN2606L0803	ACCAGACAAC-CCTAGTTCCT	1	14.98	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			2	17.20	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
i1077	TN2606L0804	GAGAGGATAT-CCCATTTCAA	1	44.73	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			2	43.45	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
i1077	TN2606L0805	ATGACGTCGC-ATCCTGACCT	1	18.54	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			2	20.52	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
i1077	TN2606L0806	CCCGTTCTCG-CCAATCCGTC	1	32.93	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			2	32.67	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
i1077	TN2606L0807	CTCTAGCGAG-GATGAAGATA	1	22.58	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			2	23.92	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
i1080	TN2606L0817	CGTCAAGGGC-GAGTGACCTA	1	14.50	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
			2	15.13	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	
i1103	TN2606L1122	TGGTTCGGGT-CTCCTGCCAC	1	18.06	/rawread03/bi_system/demultiplexing/20260619_LH00670_0340_B2522VMLT4/20260621173008/28_10_10_90_0/	

Demultiplexing - info

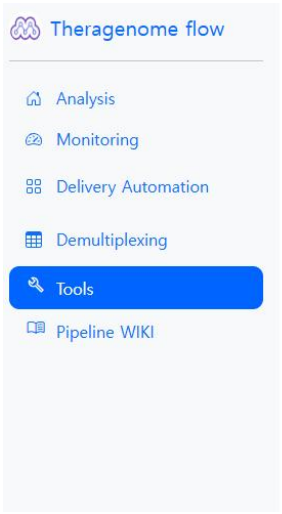








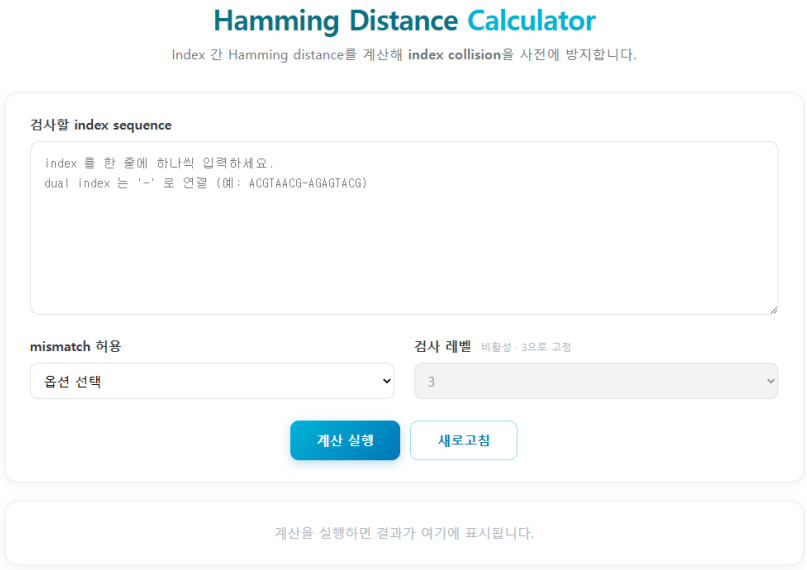
Tools

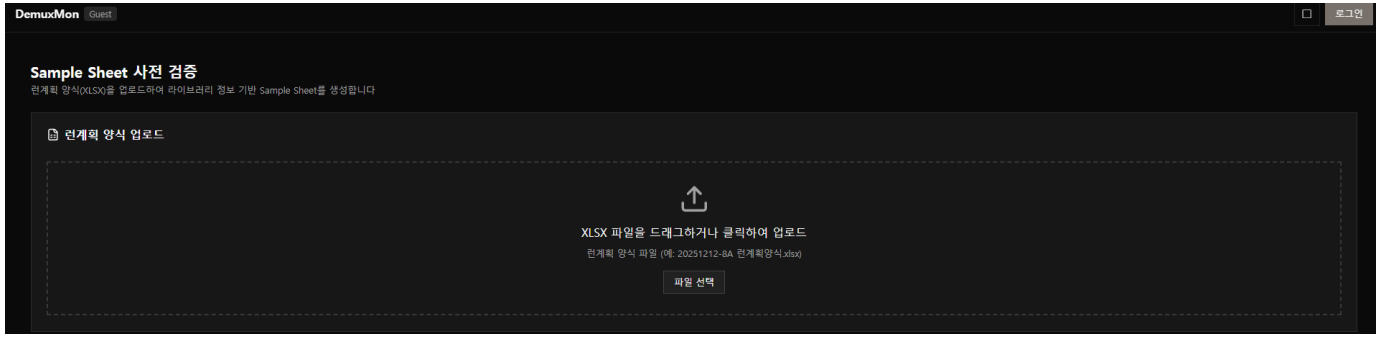




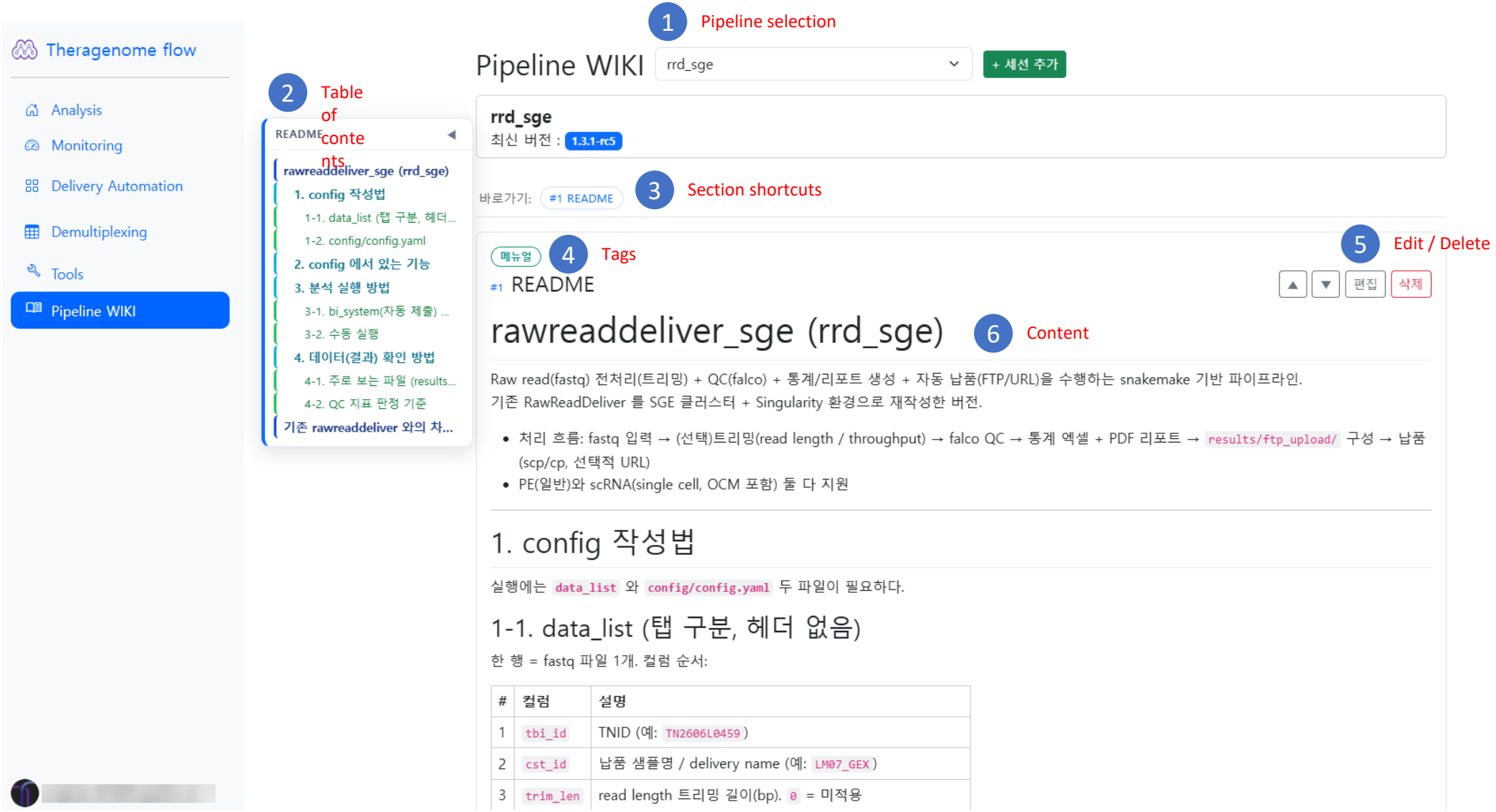
index mismatch 검사

Demultipleixng SampleSheet 검증

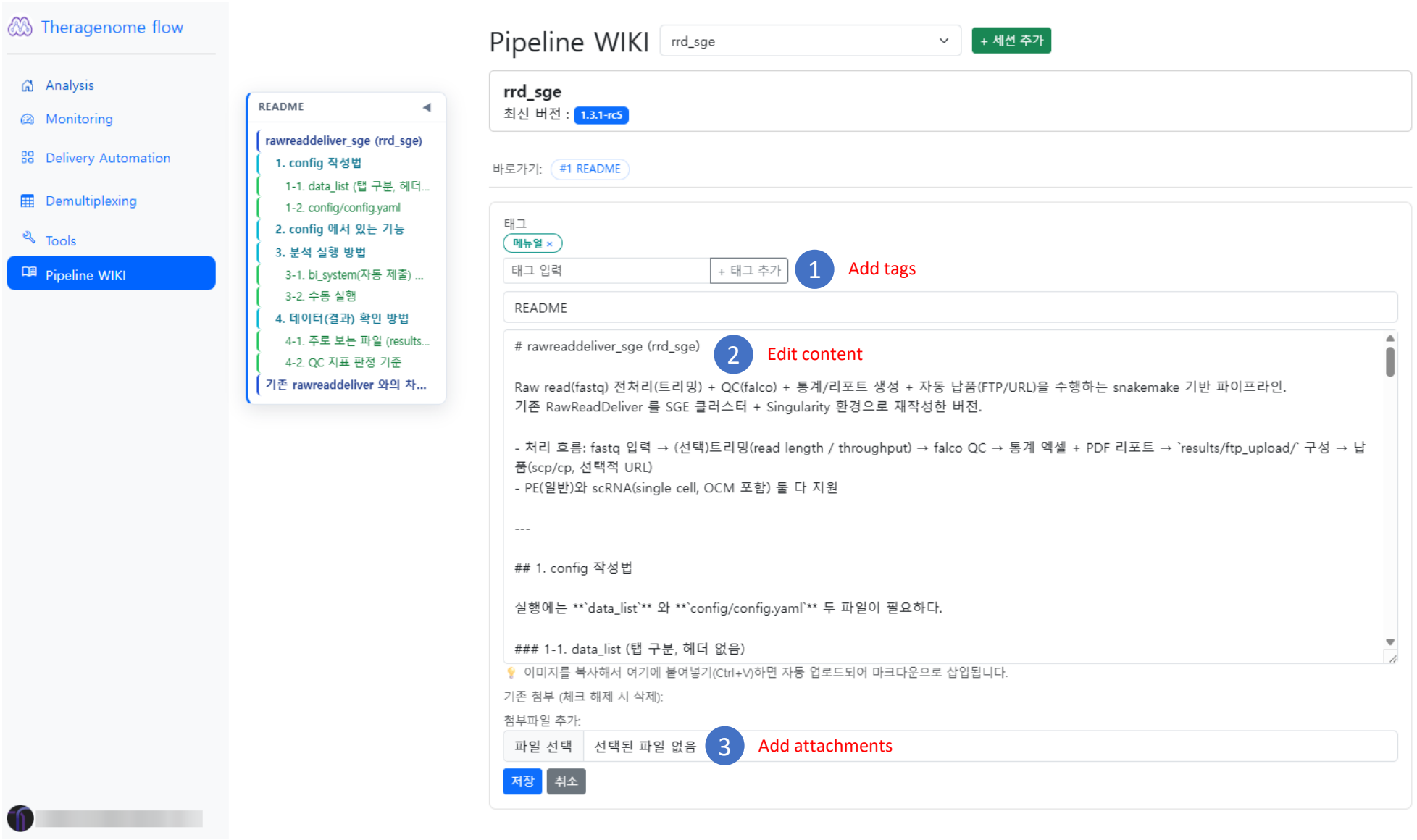




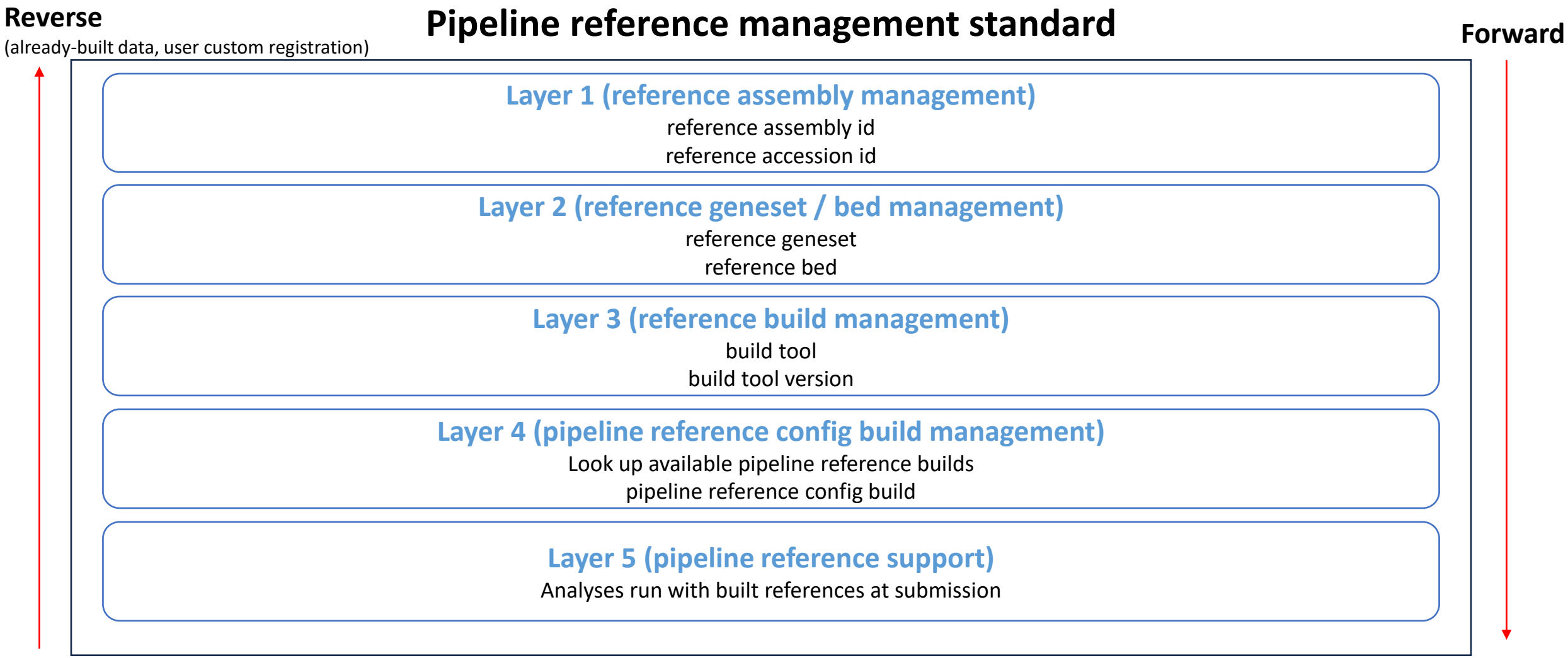
Pipeline WIKI



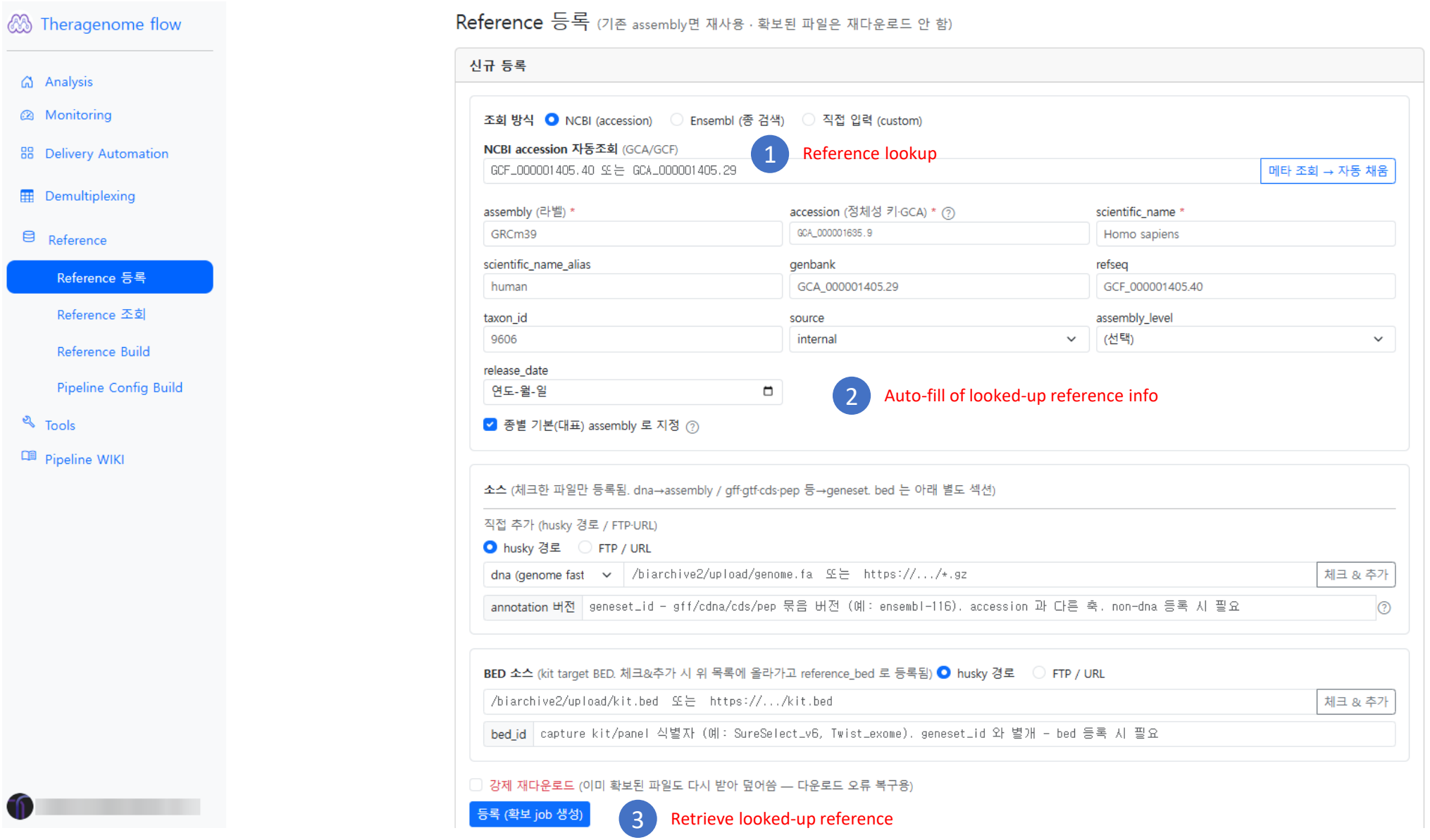
Pipeline WIKI - Edit



Centralized Reference Management - Architecture

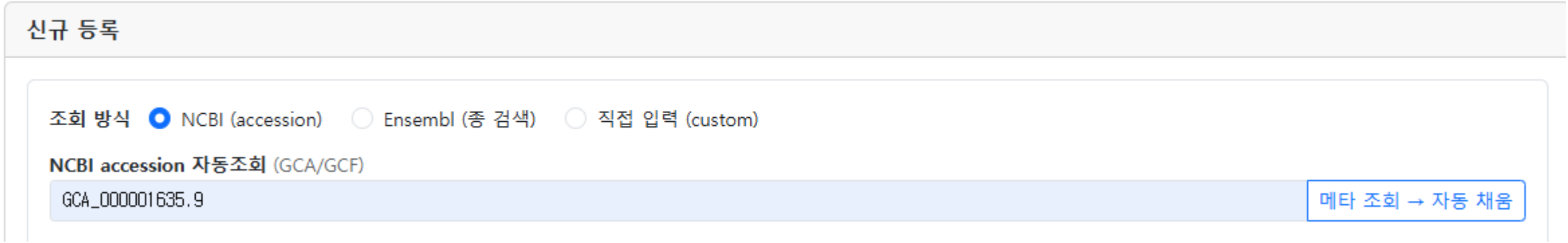


Centralized Reference Management - Registration



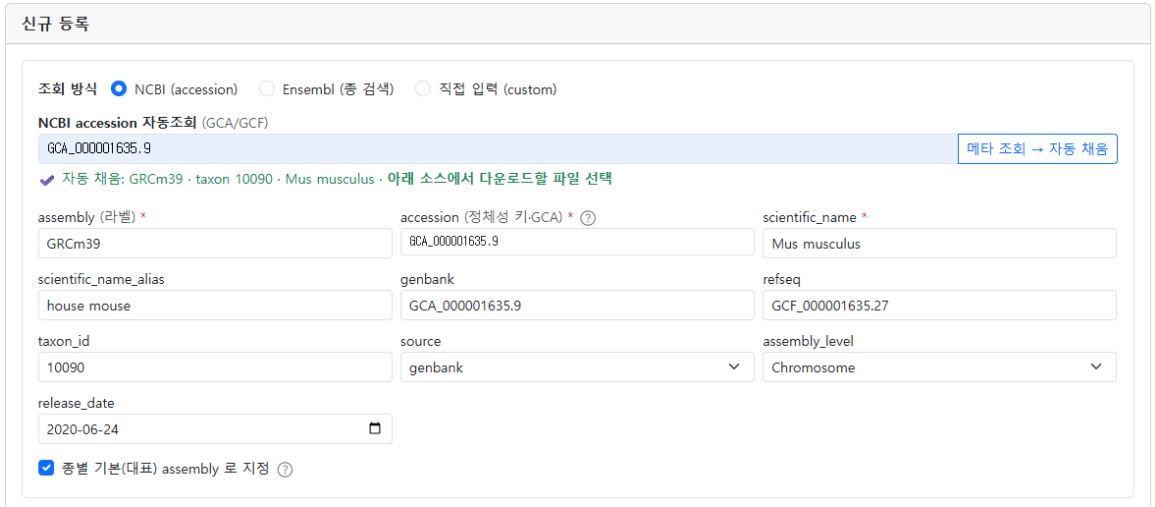
Centralized Reference Management - Registration (Example)

1 Reference lookup



Checked against the reference site and registered automatically

2 Auto-fill of looked-up reference info



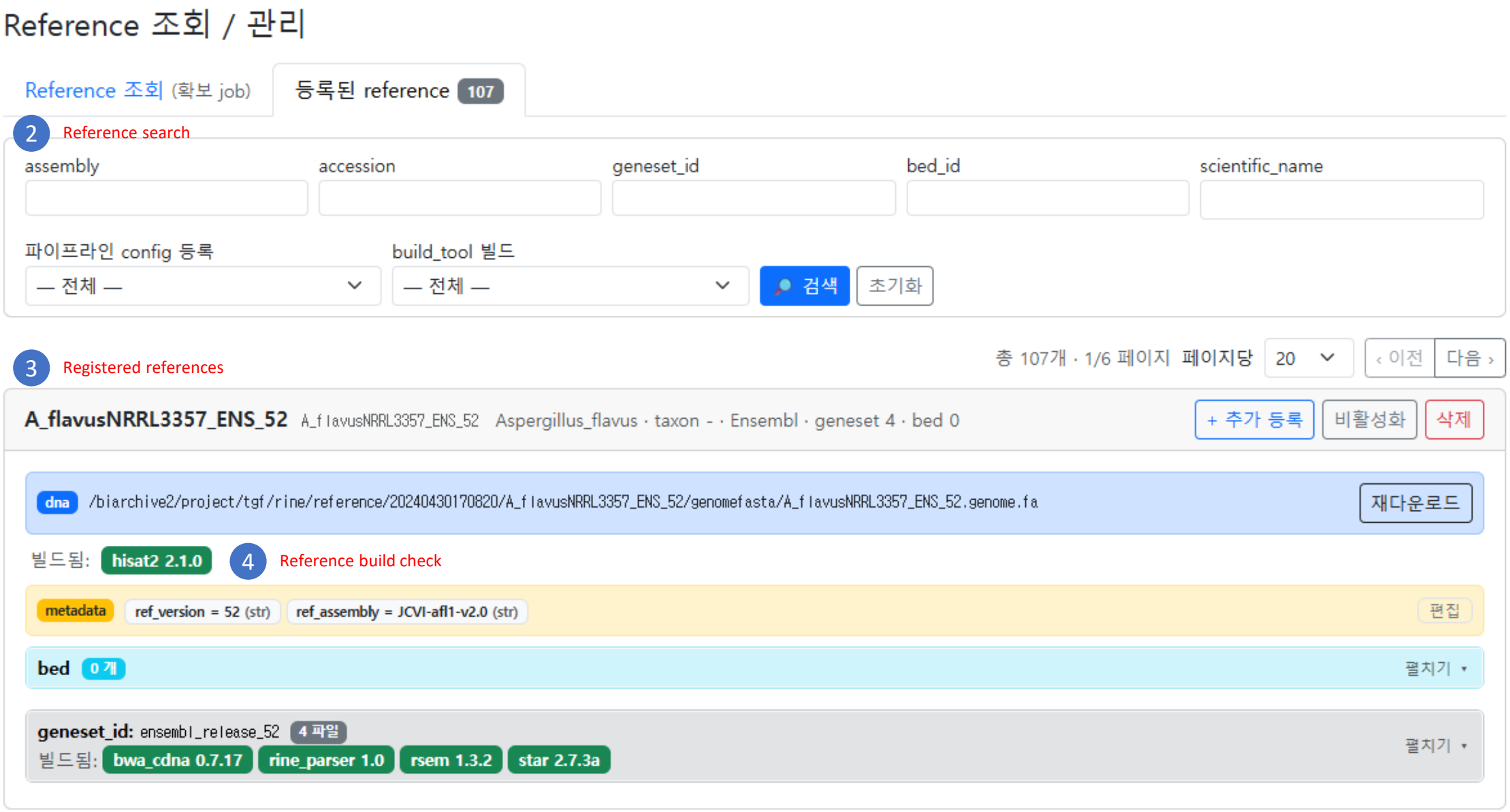
3 Retrieve looked-up reference



Centralized Reference Management - Lookup

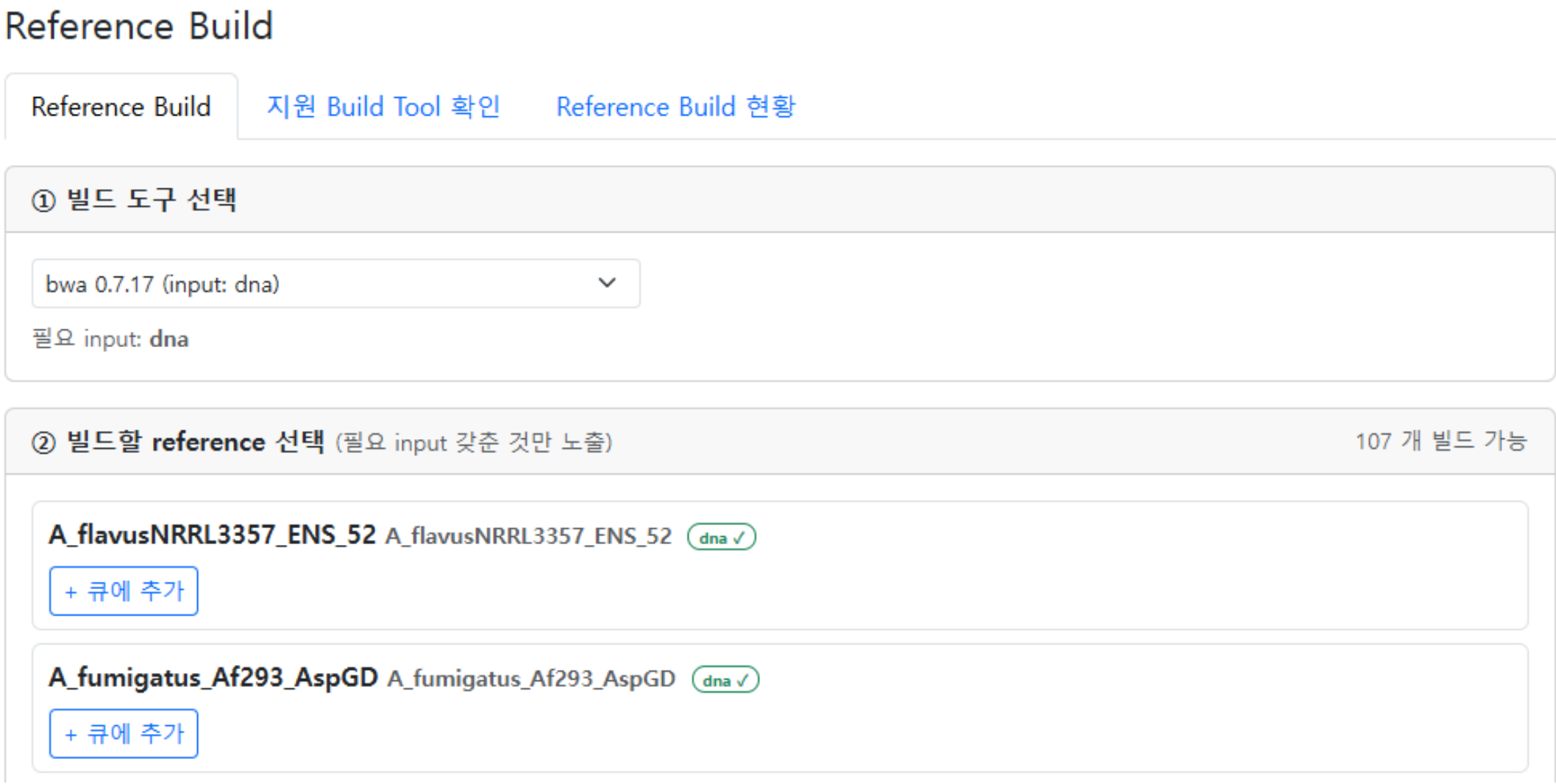


Centralized Reference Management - Lookup



Centralized Reference Management - Build and Pipeline Linking

1 Build support for individual reference tools

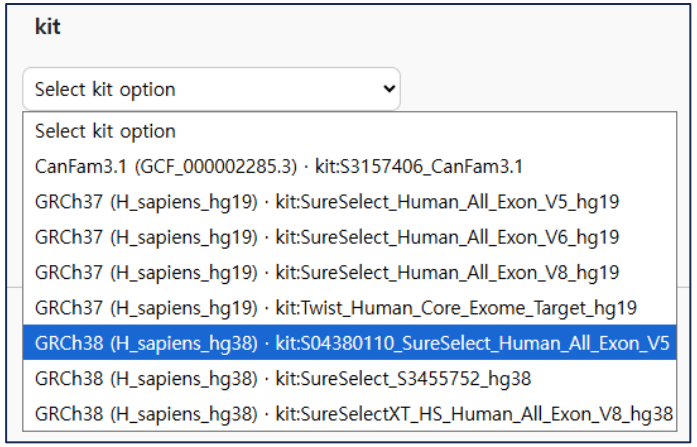


Centralized Reference Management - Build and Pipeline Linking

2 Link reference build data to pipelines (builds automatically and links to the pipeline, no separate per-reference build needed)

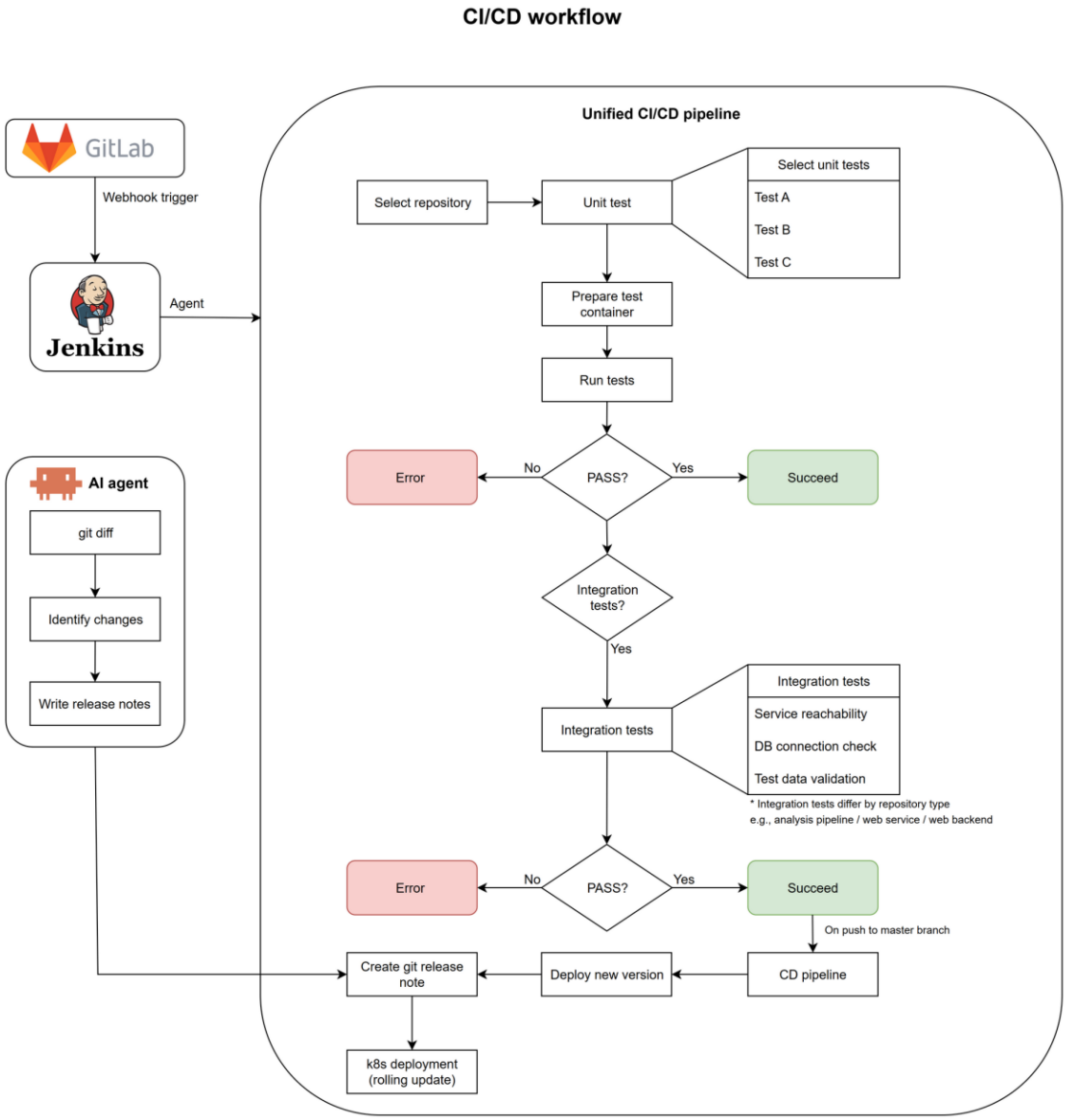


3 Automatic linking of the reference to the pipeline

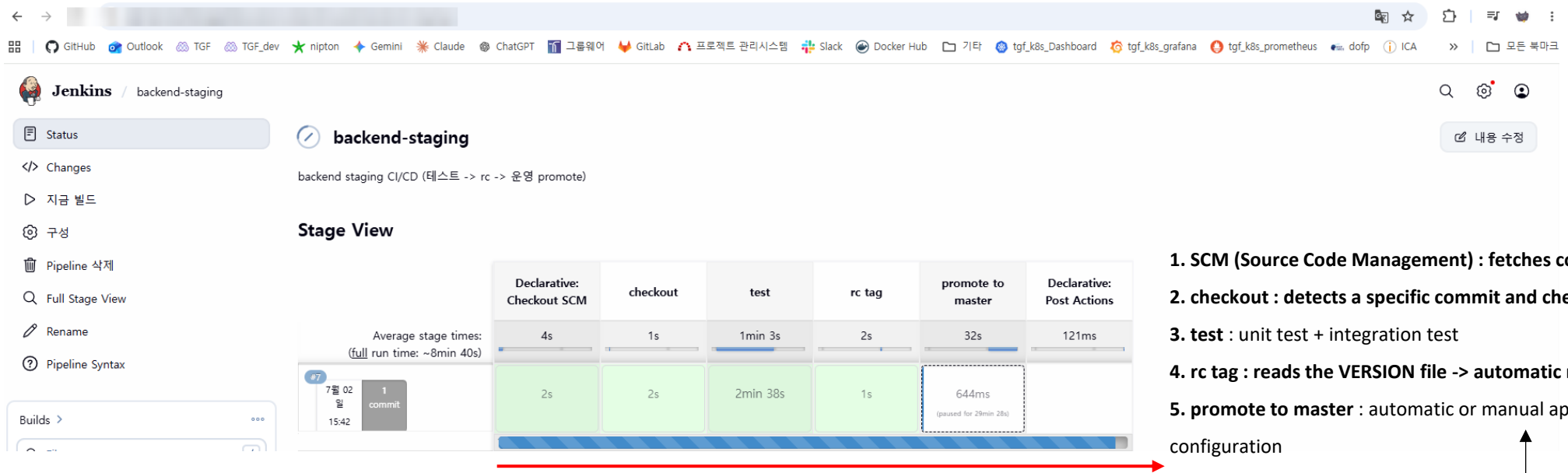


project - CI/CD pipeline

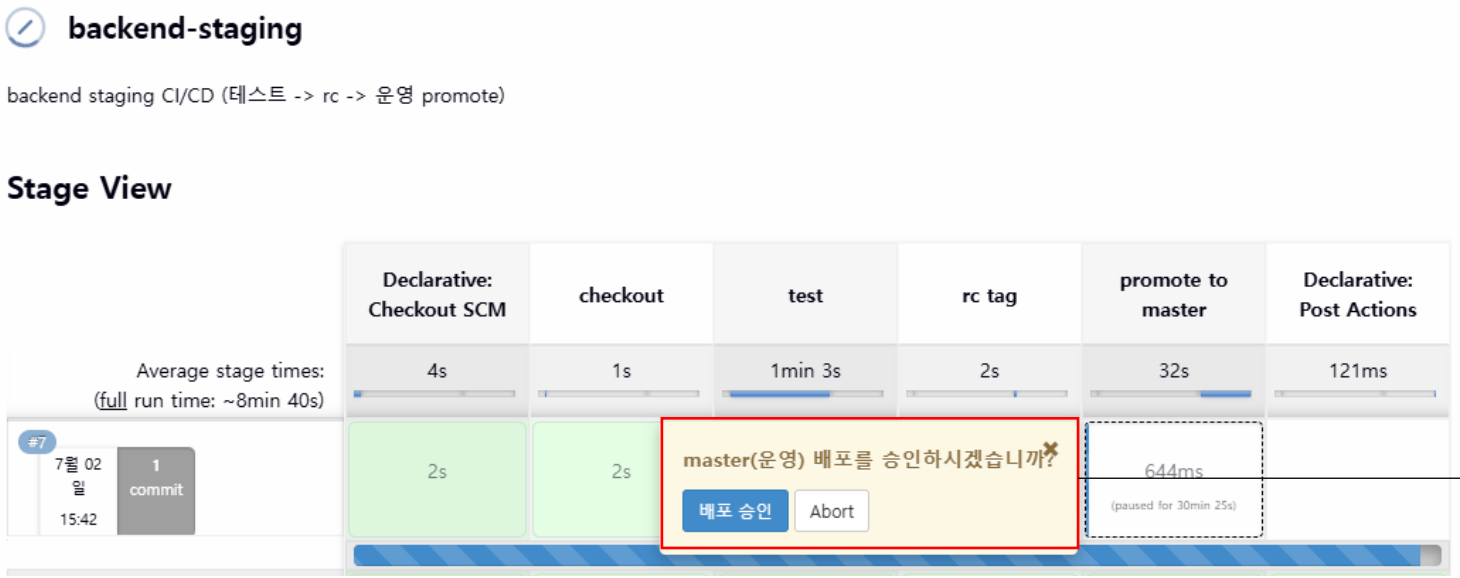
GitOps - CI/CD workflow



GitOps - Jenkins web



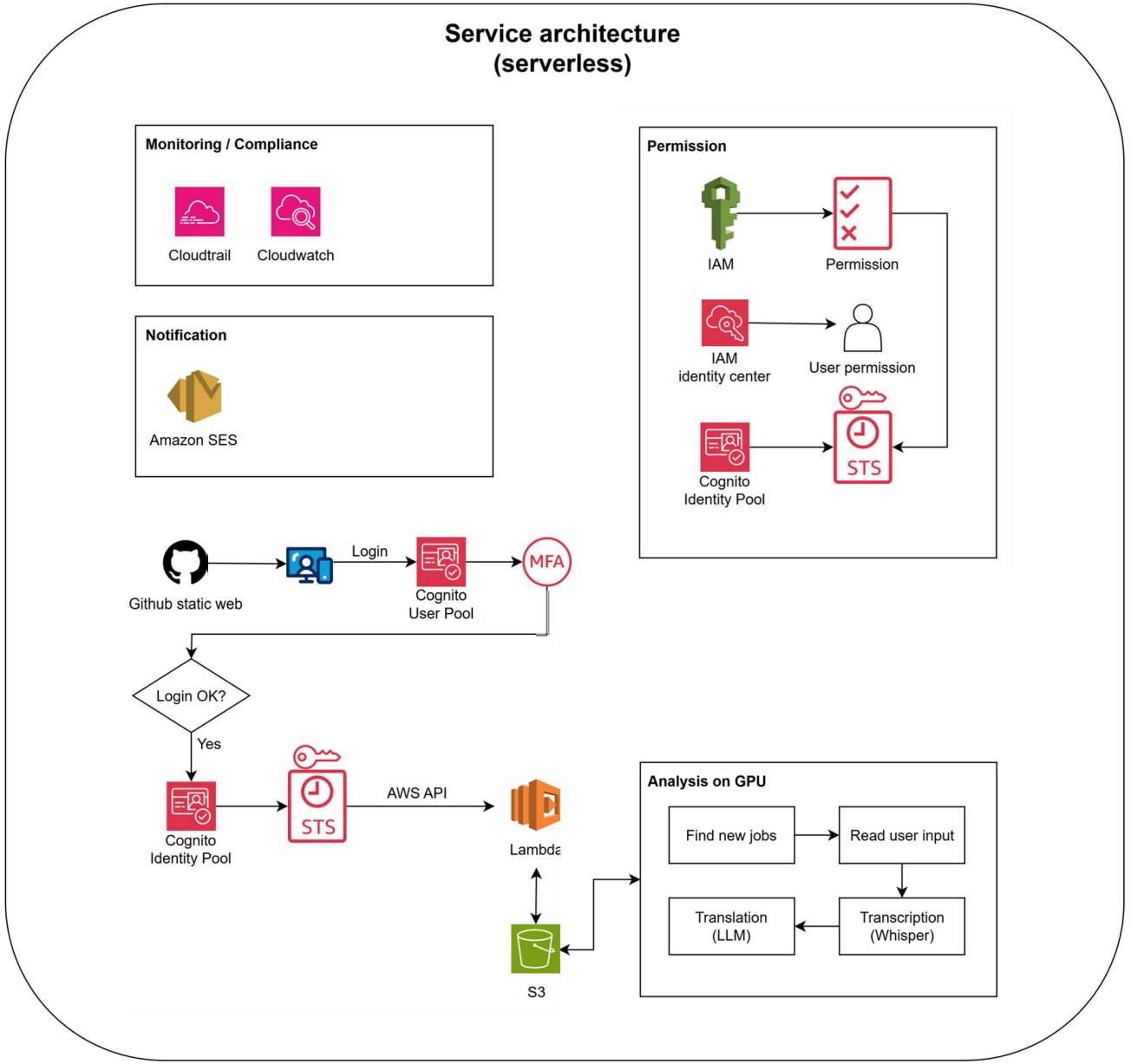
1. **SCM (Source Code Management) : fetches code from git**
2. **checkout : detects a specific commit and checks it out**
3. **test** : unit test + integration test
4. **rc tag : reads the VERSION file -> automatic rc tagging**
5. **promote to master** : automatic or manual approval, per configuration




```
jhshin@tgf-service:/d
/tools/cicd/core
$ cat config.yaml
# GitLab webhook 등록
#   글로벌 설정 (최상단
# ── 글로벌 ──
jenkins_base: "https:
secret_token: "jhshin
auto_approve: false
```

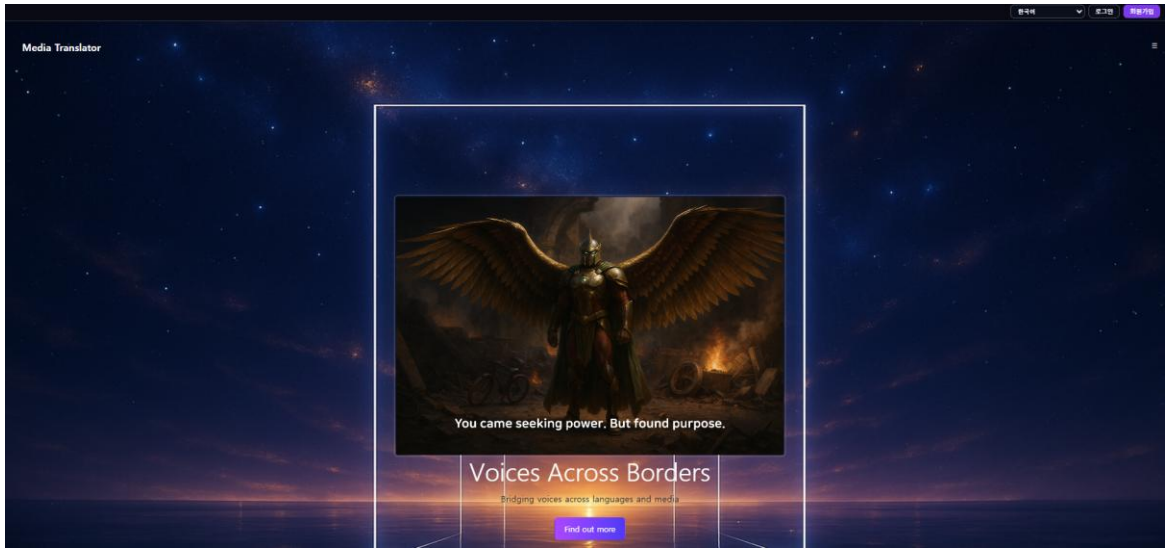
Personal project - Cognimosyne

Service Architecture



https://cognimosyne.com/

Website - landing page



media translator가 추구하는 가치

우리는 누구나 손쉽게 고품질 자막과 번역을 제작할 수 있는 세상을 꿈꿉니다.
콘텐츠 제작자, 교육자, 비즈니스 사용자가 글로벌 시청자에게 더욱 쉽고 정확하게 도달하도록 돕습니다.

🎯 목적 중심 자동화
기술 구현을 넘어 콘텐츠 본질에 집중한 자동화 기능을 제공합니다.

🌏 글로벌 확장
완전히 현지화된 자막 경험으로 전 세계 어디든 메시지를 전달합니다.

📈 SEO 자동화
검색엔진이 좋아하는 메타데이터와 키워드를 자동으로 제안해 노출을 높입니다.

🌐 다국어 자막
하나의 워크플로로 여러 언어 자막을 생성해 모든 시장을 동시에 공략하세요.

우리가 제공하는 핵심 기능

영상과 오디오에서 자막을 자동 추출하고, 문맥을 이해하는 AI로 교정한 뒤 필요한 언어로 번역합니다.
SEO 최적화를 위한 키워드와 메타데이터도 자동으로 생성되고, 직관적인 웹 기반 편집기에서 결과물을 다듬을 수 있습니다.

자막 추출
모든 영상과 오디오에서 정확한 자막을 추출해 접근성과 편집 효율을 높여줍니다.

AI 교정
AI 언어 분석으로 자막 문장을 자연스럽고 정확하게 다듬습니다.

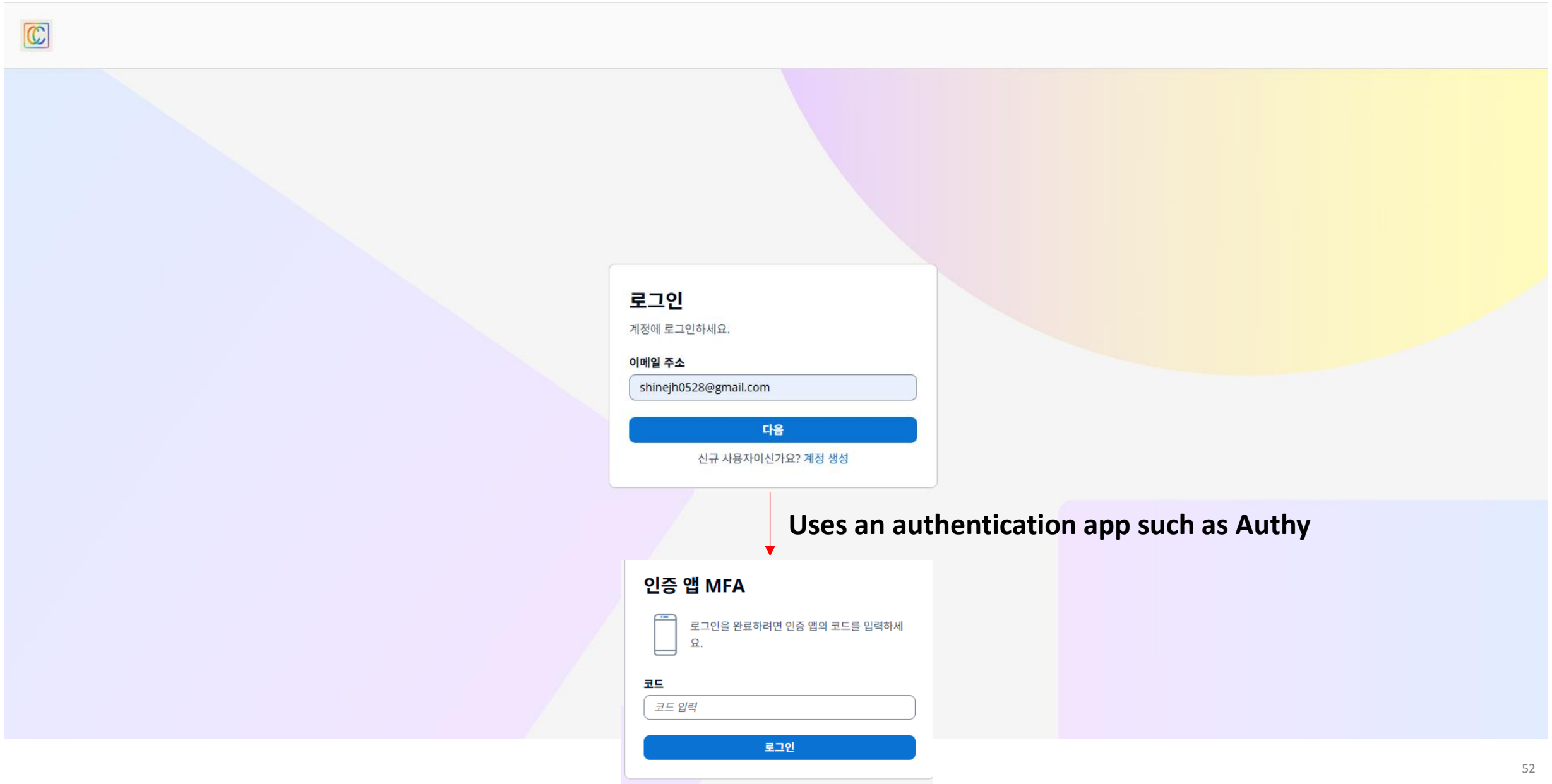
다국어 번역
한 번의 업로드로 여러 언어의 자막을 자동 생성해 글로벌 시청자에게 바로 도달합니다.

SEO 키워드 추출
콘텐츠 맥락에 맞는 키워드와 메타데이터를 만들어 검색 노출을 극대화합니다.

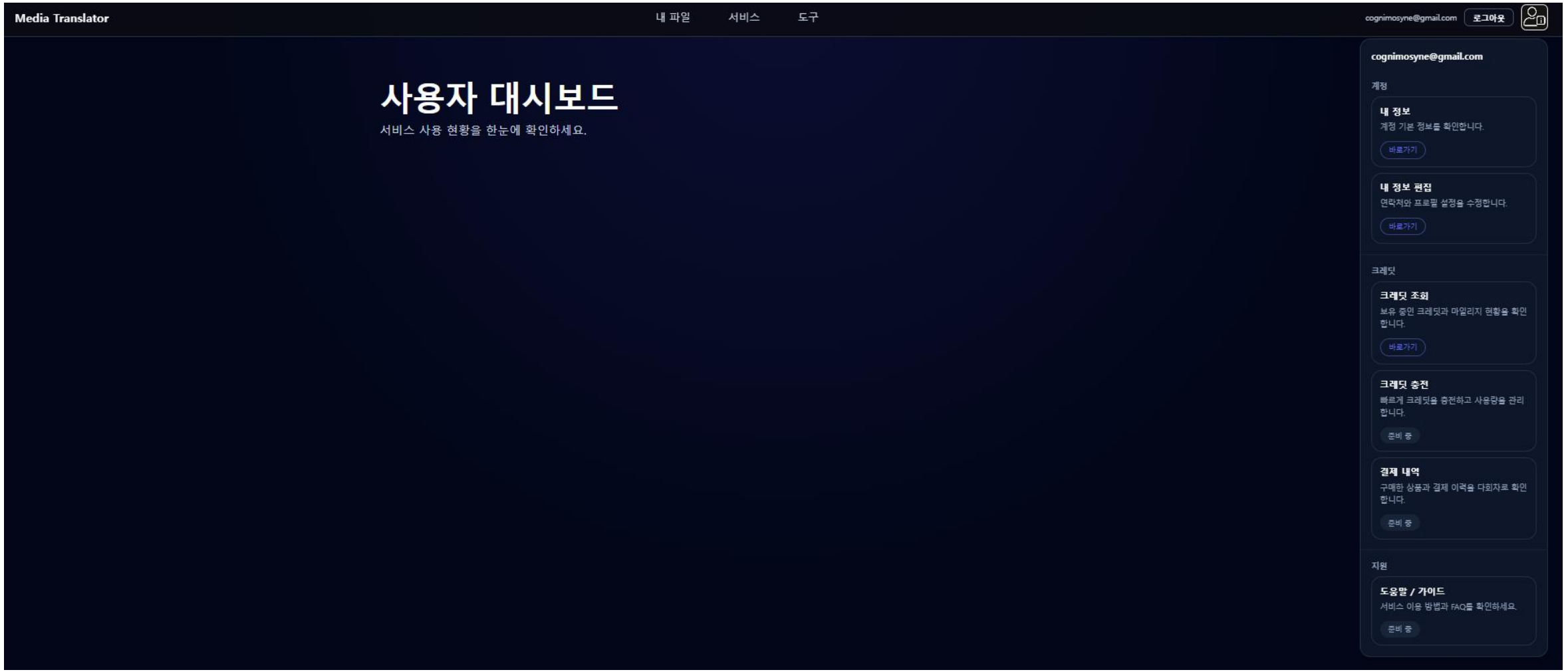
타임라인 편집
프레임 단위로 타이밍을 조정하고 텍스트도 빠르게 수정할 수 있습니다.

유연한 내보내기
완성된 자막을 SRT나 VTT 등 다양한 형식으로 내려받아 배포 채널에 바로 적용하세요.

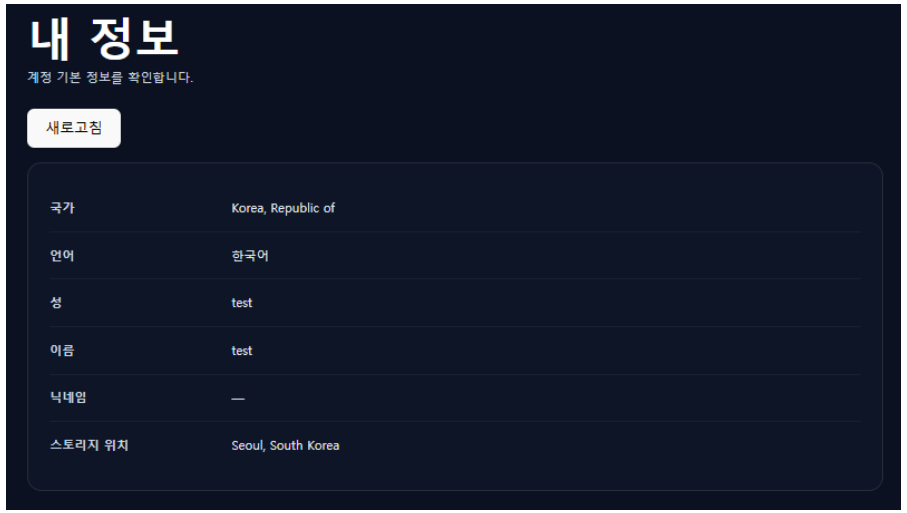
Website - login (AWS cognito)

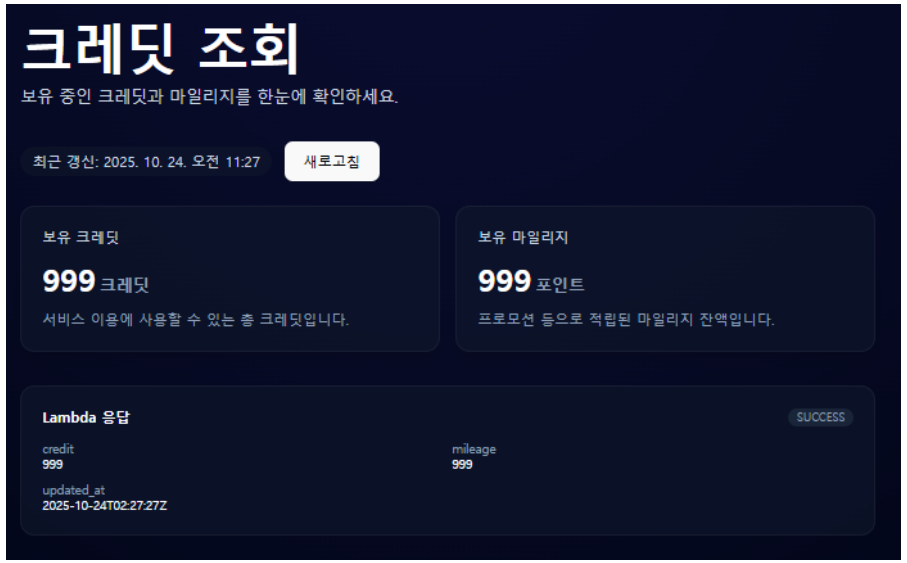


Website - User Dashboard



Website - Serverless web flow with S3 and Lambda







Thank you ! :)

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